

Artificial Neural Networks (work in progress draft)
Getting Up To Speed With The Math From Scratch

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1 Introduction

I decided to brush up my knowledge on the math behind artificial neural networks, and these are the notes that I've taken along the way.

The appendix contains a collection of notations, definitions, and theorems from the fields of linear algebra and calculus that are used throughout this document.

2 Neurons and Neural Networks

2.1 Neuron

An artificial neuron is a mathematical model that captures the behaviour of a real, biological brain cell called a "neuron".

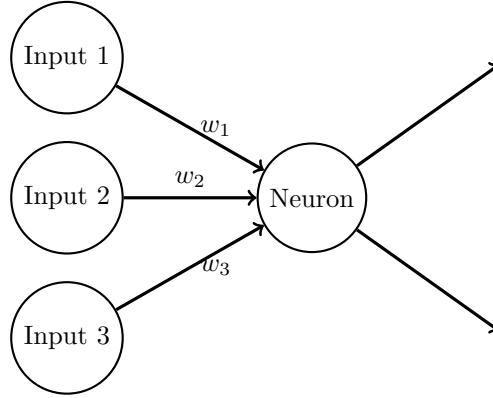


Figure 1: A neuron with 3 input connections with various weights, and 2 output connections.

A real neuron may have incoming connections from other neurons and sensing organs, and outgoing connections to other neurons, muscles, etc. Depending on its inner state and the signals that it receives from its inputs, a neuron may or may not send a signal to its outputs, and if it does send a signal, the strength of that can also vary. The strength of the outgoing signal is called the "activation" level of the neuron.

A mathematical neuron's activation is represented by a number, $a \in \mathbb{R}$.

For a neuron with no incoming connections (called an "input" neuron, also known as "feature"), the activation is determined by the raw input data, for example, a single pixel's luminosity level in photo, scaled to the $[0, 1] \subset \mathbb{R}$ interval.

For neurons with $n \in \mathbb{N}^+$ incoming connections, the activation is calculated as a function of the sum of the neuron's own bias parameter $b \in \mathbb{R}$ and the weighted sum of the activations of all the neurons from which an incoming connection to this neuron exists, with some function $f : \mathbb{R} \rightarrow \mathbb{R}$ called the "activation function" ($\mathbf{a}, \mathbf{w} \in \mathbb{R}^n$):

$$a = f \left(b + \sum_{k=0}^n w_k \cdot a_k \right) \quad (1)$$

The weights and biases for a collection of neurons are usually calculated by a training algorithm, which, based on the derivative of the activation function, will arrange the parameter values so that for a given set of input activations, the neurons will produce the desired output activations.

2.1.1 Common activation functions and their derivatives

Depending on the problem to solve, there are various activation functions to choose from. With $\beta, \gamma \in \mathbb{R}$ (either of which may be a constant or a parameter that is learned along with the weights and biases), the options include:

2.1.1.1 Threshold function

$$\text{Threshold}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

Note: $\text{Threshold}(x)$ is not differentiable at $x = 0$, and its derivative is 0 elsewhere.

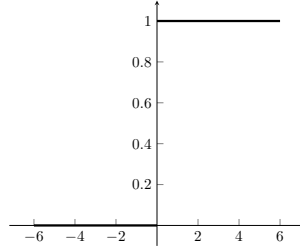


Figure 2: Plot of the $\text{Threshold}(x)$ function.

2.1.1.2 Sign function

$$\text{Sign}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0 \end{cases}$$

Note: $\text{Sign}(x)$ is not differentiable at $x = 0$, and its derivative is 0 elsewhere.

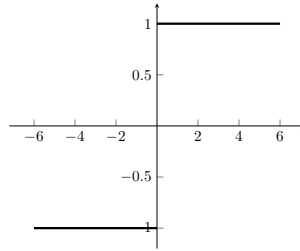


Figure 3: Plot of the $\text{Sign}(x)$ function.

2.1.1.3 Sigmoid function Also known as Logistic Curve. β is called the "steepness", and $\gamma \neq 0$ (usually $\gamma = 1$):

$$\sigma_{\beta,\gamma}(x) = \frac{\gamma}{1 + e^{-\beta \cdot x}} \quad \sigma'_{\beta,\gamma}(x) = \beta \cdot \sigma_{\beta,\gamma}(x) \cdot \left(1 - \frac{1}{\gamma} \cdot \sigma_{\beta,\gamma}(x)\right)$$

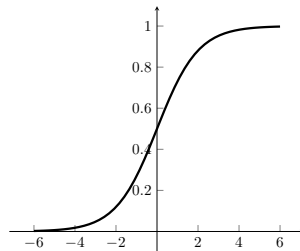


Figure 4: Plot of the $\sigma_{1,1}(x)$ function.

Note: calculating the derivative involves a few tricks. First the function is rewritten using the fact that for all $\beta \in \mathbb{R}$, the $e^{\beta \cdot x} > 0$ inequality holds:

$$\sigma_{\beta,\gamma}(x) = \frac{\gamma}{1 + e^{-\beta \cdot x}} = \frac{\gamma}{1 + \frac{1}{e^{\beta \cdot x}}} = \frac{\gamma}{1 + \frac{1}{e^{\beta \cdot x}}} \cdot 1 = \frac{\gamma}{1 + \frac{1}{e^{\beta \cdot x}}} \cdot \frac{e^{\beta \cdot x}}{e^{\beta \cdot x}} = \frac{\gamma \cdot e^{\beta \cdot x}}{e^{\beta \cdot x} + 1}$$

Then the derivative:

$$\begin{aligned} \sigma'_{\beta,\gamma}(x) &= \frac{(\gamma \cdot e^{\beta \cdot x})' \cdot (e^{\beta \cdot x} + 1) - \gamma \cdot e^{\beta \cdot x} \cdot (e^{\beta \cdot x} + 1)'}{(e^{\beta \cdot x} + 1)^2} \\ &= \frac{(\gamma \cdot \beta \cdot e^{\beta \cdot x}) \cdot (e^{\beta \cdot x} + 1) - \gamma \cdot e^{\beta \cdot x} \cdot (\beta \cdot e^{\beta \cdot x})}{(e^{\beta \cdot x} + 1)^2} \\ &= \frac{\gamma \cdot \beta \cdot e^{2 \cdot \beta \cdot x} + \gamma \cdot \beta \cdot e^{\beta \cdot x} - \gamma \cdot \beta \cdot e^{2 \cdot \beta \cdot x}}{(e^{\beta \cdot x} + 1)^2} \\ &= \frac{\gamma \cdot \beta \cdot e^{\beta \cdot x}}{(e^{\beta \cdot x} + 1)^2} \\ &= \frac{\gamma \cdot \beta \cdot e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \cdot \frac{1}{e^{\beta \cdot x} + 1} \\ &= \frac{\gamma \cdot \beta \cdot e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \cdot \frac{e^{\beta \cdot x} + 1 - e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \\ &= \frac{\gamma \cdot \beta \cdot e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \cdot \left(\frac{e^{\beta \cdot x} + 1}{e^{\beta \cdot x} + 1} - \frac{e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \right) \\ &= \frac{\gamma \cdot \beta \cdot e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \cdot \left(1 - \frac{e^{\beta \cdot x}}{e^{\beta \cdot x} + 1} \right) \\ &= \beta \cdot \sigma_{\beta,\gamma}(x) \cdot \left(1 - \frac{1}{\gamma} \cdot \sigma_{\beta,\gamma}(x) \right) \end{aligned}$$

2.1.1.4 ReLU function Rectified Linear Unit.

$$\text{ReLU}(x) = \max(0, x) \qquad \text{ReLU}'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Note: $\text{ReLU}(x)$ is not differentiable at $x = 0$, but when implementing a neural network, people just arbitrarily choose the value of $\text{ReLU}'(0)$ to be either 0 or 1.

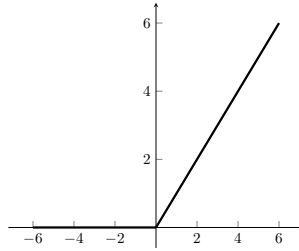


Figure 5: Plot of the $\text{ReLU}(x)$ function.

2.1.1.5 Parametric ReLU function

$$\text{PReLU}_\beta(x) = \begin{cases} x & \text{if } x > 0 \\ \beta \cdot x & \text{if } x \leq 0 \end{cases} \quad \text{PReLU}'_\beta(x) = \begin{cases} 1 & \text{if } x > 0 \\ \beta & \text{if } x < 0 \end{cases}$$

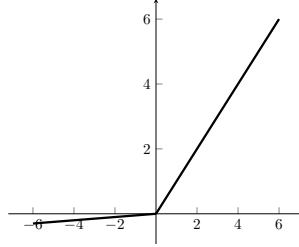


Figure 6: Plot of the $\text{PReLU}_{0.05}(x)$ function.

When β is chosen to be a small positive number, e.g. 0.01, then PReLU is also called "leaky ReLU". A leaky ReLU can help mitigating the "dying ReLU" problem (a form of the "vanishing gradient problem"), which arises when a ReLU neuron is pushed into a state in which it becomes inactive for almost all inputs, so the training algorithm will no longer be able to get it out from that state.

Note: $\text{PReLU}(x)$ is not differentiable at $x = 0$, but when implementing a neural network, people just arbitrarily chose the value of $\text{PReLU}'(0)$ to be either β or 1.

2.1.1.6 SiLU function Sigmoid Linear Unit, also known as Swish function.

$$\text{SiLU}_{\beta,\gamma}(x) = x \cdot \sigma_{\beta,\gamma}(x) \quad \text{SiLU}'_{\beta,\gamma}(x) = \sigma_{\beta,\gamma}(x) \cdot \left(1 + x - \frac{1}{\gamma} \cdot \sigma_{\beta,\gamma}(x)\right)$$

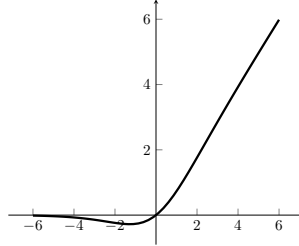


Figure 7: Plot of the $\text{SiLU}_{1,1}(x)$ function.

2.1.1.7 ELU function Exponential Linear Unit. For $0 \leq \beta$:

$$\text{ELU}_\beta(x) = \begin{cases} x & \text{if } x > 0 \\ \beta \cdot (e^x - 1) & \text{if } x \leq 0 \end{cases} \quad \text{ELU}'_\beta(x) = \begin{cases} 1 & \text{if } x > 0 \\ \beta \cdot e^x & \text{if } x \leq 0 \end{cases}$$

Note: strictly speaking, when $\beta \neq 1$, then $\text{ELU}_\beta(x)$ is not differentiable at $x = 0$.

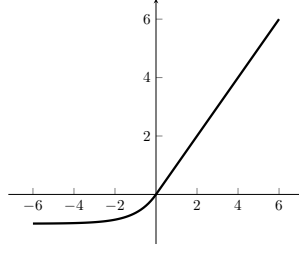


Figure 8: Plot of the $\text{ELU}_1(x)$ function.

2.1.1.8 Softplus function Also known as SmoothReLU function. β is called the "sharpness":

$$\text{Softplus}_{\beta,\gamma}(x) = \frac{\gamma}{\beta} \cdot \ln(1 + e^{\beta \cdot x}) \quad \text{Softplus}'_{\beta,\gamma}(x) = \sigma_{\beta,\gamma}(x)$$

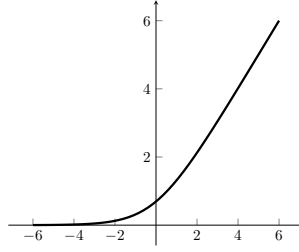


Figure 9: Plot of the $\text{Softplus}_{1,1}(x)$ function.

2.1.1.9 Mish function

$$\begin{aligned} \text{Mish}_{\beta}(x) &= x \cdot \tanh(\text{Softplus}_{\beta,1}(x)) \\ \text{Mish}'_{\beta}(x) &= \tanh(\text{Softplus}_{\beta,1}(x)) + x \cdot \sigma_{\beta,1}(x) \cdot \text{sech}^2(\text{Softplus}_{\beta,1}(x)) \end{aligned}$$

Where the $\tanh : \mathbb{R} \rightarrow \mathbb{R}$ and the $\text{sech} : \mathbb{R} \rightarrow \mathbb{R}$ functions can be defined as:

$$\tanh(x) = \frac{e^{2 \cdot x} - 1}{e^{2 \cdot x} + 1} \quad \text{sech}(x) = \frac{2}{e^x + e^{-x}}$$

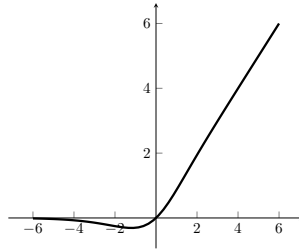


Figure 10: Plot of the $\text{Mish}_1(x)$ function.

2.1.1.10 Squareplus function For $0 \leq \beta$:

$$\text{Squareplus}_{\beta}(x) = \frac{x + \sqrt{x^2 + \beta}}{2}$$

$$\text{Squareplus}'_{\beta}(x) = \frac{1}{2} + \frac{x}{2 \cdot \sqrt{x^2 + \beta}}$$

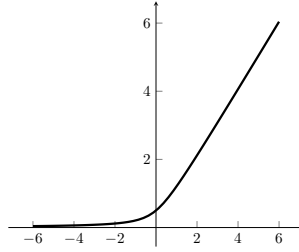


Figure 11: Plot of the $\text{Squareplus}_1(x)$ function.

2.2 Neural network

Neurons are arranged in layers, the activations of the neurons in a layer are the inputs of the next layer.

Let $L \in \mathbb{N}^+$ denote the number of non-input layers in a neural network.

Let $\mathbf{n} \in (\mathbb{N}^+)^L$ denote the number of neurons in each layer of the network, and let $n_0 \in \mathbb{N}^+$ denote the number of input neurons (features).

For $L \geq k \in \mathbb{N}$, the $\mathbf{a}^{(k)} \in \mathbb{R}^{n_k}$ vector represents the activations in the k -th layer; $\mathbf{a}^{(0)}$ is the input layer, $\mathbf{a}^{(L)}$ is the output layer. Layers between the input and the output layer are called "hidden layers". (The parenthesized superscript here is used for indexing the layers, and not for powers.)

The interpretation of the output layer depends on the problem that the network is supposed to solve, and it may require extensive experimentation to see what works best. For example, a network that recognizes handwritten digits may produce its output in a single neuron, where a value in the $[0, 0.1]$ interval means 0, a value in the $(0.1, 0.2]$ interval means 1, and so on, or it could produce its output in 10 different neurons where the first one shows how much probability the network assigns to the input image being a handwritten 0, the second one showing the probability of a handwritten 1, and so on. (There could even be an 11th neuron which signals an unrecognized character.)

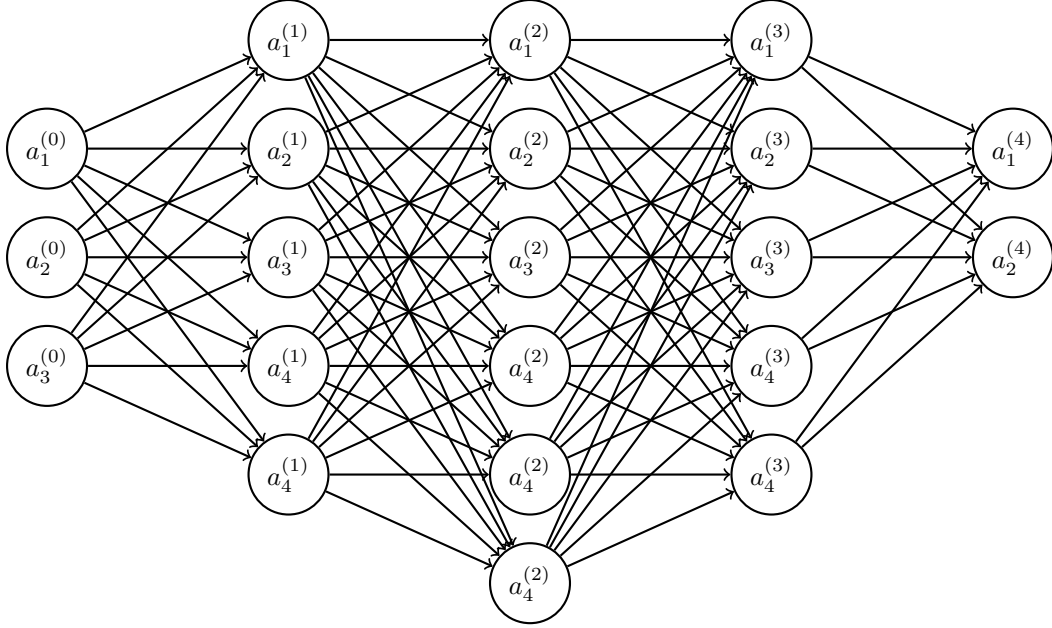


Figure 12: A neural network with 3 input neurons, 3 hidden layers, and 2 output neurons.

For $L \geq k \in \mathbb{N}^+$, the $\mathbf{W}^{(k)} \in \mathbb{R}^{n_k \times n_{k-1}}$ matrix represents the weights of the connections from the $k-1$ -th layer to the k -th layer, and the $\mathbf{b}^{(k)} \in \mathbb{R}^{n_k}$ vector represents the biases in the k -th layer.

Let $\mathbf{f}^{(k)}(\mathbf{x})$ for $\mathbf{x} \in \mathbb{R}^{n_k}$ denote the $\left[f_j^{(k)}(x_j) \right]_{j=1}^{n_k} \in \mathbb{R}^{n_k}$ vector where $f_j^{(k)} : \mathbb{R} \rightarrow \mathbb{R}$ is the function that is used for calculating the activation of the j -th neuron in the k -th layer.

The activation of the k -th layer (where $L \geq k \in \mathbb{N}^+$) can be calculated as:

$$\mathbf{a}^{(k)} = \mathbf{f}^{(k)} \left(\mathbf{W}^{(k)} \cdot \mathbf{a}^{(k-1)} + \mathbf{b}^{(k)} \right) \quad (2)$$

2.2.1 Simplified notation

For $L \geq k \in \mathbb{N}$, let $a_0^{(k)} = 1$, and if $k > 0$ then for $n_k \geq j \in \mathbb{N}^+$, let $w_{j,0}^{(k)} = b_j^{(k)}$, and for $n_{k-1} \geq i \in \mathbb{N}$, let $w_{0,i} = 1$, and let $f_0^{(k)} = \text{id}$, and let $w_{0,0} = 0$.

With this notation, equation 2 can be simplified:

$$\mathbf{a}^{(k)} = \mathbf{f}^{(k)} \left(\mathbf{W}^{(k)} \cdot \mathbf{a}^{(k-1)} \right) \quad (3)$$

2.2.2 Examples

2.2.2.1 Feed-forward network A network with no loops (backward connections).

2.2.2.2 Recurrent network A network with loops (backward connections).

2.2.2.3 Perceptron A single layer, single output ($L = 1, n_1 = 1$) feed-forward network.

2.2.2.3.1 Majority function perceptron Tells if more than half of its inputs are 1. $w_{1,j} = 1$ ($n_0 \geq j \in \mathbb{N}^+$) and $b_1 = \frac{n_0}{2}$.

2.2.2.3.2 Linear separator With the simplified notation (2.2.1), the activation of the perceptron can be written as:

$$a^{(1)} = f \left(\mathbf{W}^{(1)} \cdot \mathbf{a}^{(0)} \right) \quad (4)$$

Since the $\mathbf{W}^{(1)} \cdot \mathbf{a}^{(0)} = 0$ equation defines a hyperplane in n_0 dimensions, a perceptron with f being the threshold function (2.1.1.1) can be thought of as a classifier which separates inputs that lie on one side of the plane from the inputs that lie on the other side.

A binary function ($f : \mathbb{R}^n \rightarrow \{0, 1\}$ for some $n \in \mathbb{N}^+$) is linearly separable if its inputs that yield 1 can be separated from the inputs which yield 0 with a straight line (or plane, or hyperplane). For example, the OR, AND : $\{0, 1\}^2 \rightarrow \{0, 1\}$ functions are linearly separable:

$$\begin{aligned} \text{OR}(x, y) &= \begin{cases} 0 & \text{if } x = y = 0 \\ 1 & \text{otherwise} \end{cases} \\ \text{AND}(x, y) &= \begin{cases} 1 & \text{if } x = y = 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

But the XOR : $\{0, 1\}^2 \rightarrow \{0, 1\}$ function is not linearly separable:

$$\text{XOR}(x, y) = \begin{cases} 1 & \text{if } (x, y) \in \{(0, 1), (1, 0)\} \\ 0 & \text{otherwise} \end{cases}$$

Note that a function which is not linearly separable in $n \in \mathbb{N}^+$ dimensions may be turned into a linearly separable function by cleverly adding extra dimensions.

3 Learning, gradient descent, backpropagation

Before the training of a neural network begins, all its weights and biases are initialized with random values. The activation of the output layer is then calculated for a number of example inputs for which the desired output is already known, and the weights and biases are adjusted so that the average difference between the calculated and the desired outputs is minimized. (This is called "supervised learning".)

The idea is to define an error function in terms of the weights and biases in the network, treating the input activations like constants. If all the activation functions are differentiable, then this error function is also differentiable, so its gradient for a given set of weights and biases will tell the direction and the rate of the fastest increase. Since the goal is to minimize the error, we will multiply the gradient by -1 to get the direction of the fastest decrease, and use that to adjust the weights and biases for each example input. As more and more adjustments are made for more and more examples, the error function will hopefully converge towards a (local) minimum and thus, its gradient should converge towards 0. (Hence the name "gradient descent".)

In some specific cases (e.g. convex functions), guaranteed convergence can be proven and its speed can be calculated, but in practice, the error function rarely has such nice properties, so experimentation tends to be more useful than attempting to produce formal theoretical proofs.

(From now on, the differentiability of $f_j^{(k)}$ is assumed for all $j, k \in \mathbb{N}^+, k \leq L, j \leq n_k$.)

3.1 Error function

(Also known as Cost function.)

Let $\mathbf{y} \in \mathbb{R}^{n_L}$ denote the expected output activations for a given $\mathbf{a}^{(0)} \in \mathbb{R}^{n_0}$ example input. One way to define an error function for this single example input is the following (where $C : \mathbb{R}^{n_0} \rightarrow \mathbb{R}$):

$$\begin{aligned} C(\mathbf{a}^{(0)}) &= \frac{1}{2} \cdot \left\| \mathbf{a}^{(L)} - \mathbf{y} \right\|_2^2 \\ &= \frac{1}{2} \cdot \left(\sqrt{\sum_{j=1}^{n_L} (a_j^{(L)} - y_j)^2} \right)^2 = \frac{1}{2} \cdot \sum_{j=1}^{n_L} (a_j^{(L)} - y_j)^2 \\ &= \sum_{j=1}^{n_L} \frac{1}{2} \cdot (a_j^{(L)} - y_j)^2 \end{aligned} \tag{5}$$

3.1.1 Gradient

To find ∇C , we need the partial derivatives for each $w_{u^{(k)}, v^{(k-1)}}^{(k)}$ (for $k, u^{(k)}, v^{(k-1)} \in \mathbb{N}, 1 \leq k \leq L, 1 \leq u^{(k)} \leq n_k, v^{(k-1)} \leq n_{k-1}$).

3.1.1.1 Notation

3.1.1.1.1 $\Delta_j^{(k)}$ For a given $L \geq k \in \mathbb{N}^+$ and $n_k \geq j \in \mathbb{N}$ and $\mathbf{a}^{(0)} \in \mathbb{R}^{n_0}$, let

$$\Delta_j^{(k)} = f_j^{(k)'} \left(\sum_{i=0}^{n_{k-1}} w_{j,i}^{(k)} \cdot a_i^{(k-1)} \right) \tag{6}$$

and

$$\Delta^{(k)} = \left[\Delta_j^{(k)} \right]_{j=0}^{n_k} = \begin{bmatrix} \Delta_0^{(k)} \\ \Delta_1^{(k)} \\ \Delta_2^{(k)} \\ \vdots \\ \Delta_{n_k}^{(k)} \end{bmatrix} \quad (7)$$

Note:

$$\Delta_0^{(k)} = f_0^{(k)'} \left(\sum_{i=0}^{n_{k-1}} w_{j,i}^{(k)} \cdot a_i^{(k-1)} \right) = \text{id}' \left(\sum_{i=0}^{n_{k-1}} w_{0,i}^{(k)} \cdot a_i^{(k-1)} \right) = 1 \quad (8)$$

3.1.1.1.2 $\mathbf{f}^{(k)'}(\mathbf{x})$ For a given $L \geq k \in \mathbb{N}^+$ and $\mathbf{x} \in \mathbb{R}^{n_k}$:

$$\mathbf{f}^{(k)'}(\mathbf{x}) = \left[f_j^{(k)'}(x_j) \right]_{j=0}^{n_k} = \begin{bmatrix} f_0^{(k)'}(x_0) \\ f_1^{(k)'}(x_1) \\ f_2^{(k)'}(x_2) \\ \vdots \\ f_{n_k}^{(k)'}(x_{n_k}) \end{bmatrix} \quad (9)$$

3.1.1.2 Output layer Partial derivatives using equation 5 ($u, v \in \mathbb{N}, 1 \leq u \leq n_L, v \leq n_{L-1}$):

$$\begin{aligned} \frac{\partial}{\partial w_{u,v}^{(L)}} C(\mathbf{a}^{(0)}) &= \frac{\partial}{\partial w_{u,v}^{(L)}} \sum_{j=1}^{n_L} \frac{1}{2} \cdot \left(a_j^{(L)} - y_j \right)^2 \\ &= \sum_{j=1}^{n_L} \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(L)}} \left(a_j^{(L)} - y_j \right)^2 \\ &= \sum_{j=1}^{n_L} \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(L)}} \left(f_j^{(L)} \left(\sum_{i=0}^{n_{L-1}} w_{j,i}^{(L)} \cdot a_i^{(L-1)} \right) - y_j \right)^2 \end{aligned} \quad (10)$$

Since most of the terms in the outer summation don't depend on $w_{u,v}^{(L)}$, they can be treated as constants, and therefore their derivative is 0, and the only one that remains is where $j = u$:

$$\frac{\partial}{\partial w_{u,v}^{(L)}} C(\mathbf{a}^{(0)}) = \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(L)}} \left(f_u^{(L)} \left(\sum_{i=0}^{n_{L-1}} w_{u,i}^{(L)} \cdot a_i^{(L-1)} \right) - y_u \right)^2 \quad (11)$$

Applying the chain rule (note that the $a_i^{(L-1)}$ terms don't depend on $w_{u,v}^{(L)}$ either):

$$\frac{\partial}{\partial w_{u,v}^{(L)}} C(\mathbf{a}^{(0)}) = \frac{1}{2} \cdot 2 \cdot \left(f_u^{(L)} \left(\sum_{i=0}^{n_{L-1}} w_{u,i}^{(L)} \cdot a_i^{(L-1)} \right) - y_u \right) \cdot f_u^{(L)'} \left(\sum_{i=0}^{n_{L-1}} w_{u,i}^{(L)} \cdot a_i^{(L-1)} \right) \cdot a_v^{(L-1)} \quad (12)$$

Rearranging and shortening:

$$\frac{\partial}{\partial w_{u,v}^{(L)}} C(\mathbf{a}^{(0)}) = \left(a_u^{(L)} - y_u \right) \cdot \Delta_u^{(L)} \cdot a_v^{(L-1)} \quad (13)$$

3.1.1.3 Last hidden layer Partial derivatives for the $L - 1$ -th layer if $L \geq 2$, for $u, v \in \mathbb{N}, 1 \leq u \leq n_{L-1}, v \leq n_{L-2}$, using equation 5:

$$\begin{aligned} \frac{\partial}{\partial w_{u,v}^{(L-1)}} C(\mathbf{a}^{(0)}) &= \frac{\partial}{\partial w_{u,v}^{(L-1)}} \sum_{j=1}^{n_L} \frac{1}{2} \cdot \left(a_j^{(L)} - y_j\right)^2 \\ &= \sum_{j=1}^{n_L} \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} \left(a_j^{(L)} - y_j\right)^2 \end{aligned} \quad (14)$$

Now all of the terms inside the summation depend on $w_{u,v}^{(L-1)}$, because $a_j^{(L)}$ depends on it for all $n_L \geq j \in \mathbb{N}^+$.

Using the chain rule, expanding $a_j^{(L)}$ for a given $n_L \geq j \in \mathbb{N}^+$, and exploiting the fact that none of the weights and biases in the output layer depend on the weights and biases in the last hidden layer:

$$\begin{aligned} \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} \left(a_j^{(L)} - y_j\right)^2 &= \frac{1}{2} \cdot 2 \cdot \left(a_j^{(L)} - y_j\right) \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} a_j^{(L)} \\ &= \left(a_j^{(L)} - y_j\right) \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} f_j^{(L)} \left(\sum_{i=0}^{n_{L-1}} w_{j,i}^{(L)} \cdot a_i^{(L-1)}\right) \\ &= \left(a_j^{(L)} - y_j\right) \cdot f_j^{(L)'} \left(\sum_{i=0}^{n_{L-1}} w_{j,i}^{(L)} \cdot a_i^{(L-1)}\right) \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} \left(\sum_{i=0}^{n_{L-1}} w_{j,i}^{(L)} \cdot a_i^{(L-1)}\right) \\ &= \left(a_j^{(L)} - y_j\right) \cdot \Delta_j^{(L)} \cdot \left(\sum_{i=0}^{n_{L-1}} w_{j,i}^{(L)} \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} a_i^{(L-1)}\right) \end{aligned} \quad (15)$$

Since only $a_u^{(L-1)}$ depends on $w_{u,v}^{(L-1)}$, most of the terms in the summation are 0 (because they are multiplied by the derivative of a constant, which is 0), and the only one that remains is where $i = u$. Continuing equation 15:

$$\frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} \left(a_j^{(L)} - y_j\right)^2 = \left(a_j^{(L)} - y_j\right) \cdot \Delta_j^{(L)} \cdot w_{j,u}^{(L)} \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} a_u^{(L-1)} \quad (16)$$

Expanding $a_u^{(L-1)}$, and using the chain rule again:

$$\begin{aligned} \frac{\partial}{\partial w_{u,v}^{(L-1)}} a_u^{(L-1)} &= \frac{\partial}{\partial w_{u,v}^{(L-1)}} f_u^{(L-1)} \left(\sum_{m=0}^{n_{L-2}} w_{u,m}^{(L-1)} \cdot a_m^{(L-2)}\right) \\ &= f_u^{(L-1)'} \left(\sum_{m=0}^{n_{L-2}} w_{u,m}^{(L-1)} \cdot a_m^{(L-2)}\right) \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} \left(\sum_{m=0}^{n_{L-2}} w_{u,m}^{(L-1)} \cdot a_m^{(L-2)}\right) \\ &= \Delta_u^{(L-1)} \cdot \left(\sum_{m=0}^{n_{L-2}} \frac{\partial}{\partial w_{u,v}^{(L-1)}} w_{u,m}^{(L-1)} \cdot a_m^{(L-2)}\right) \end{aligned} \quad (17)$$

Since only one term in the summation depends on $w_{u,v}^{(L-1)}$ (the one where $m = v$), most of the terms are 0, because they are multiplied by the derivative of a constant, which is 0. Continuing equation 17:

$$\frac{\partial}{\partial w_{u,v}^{(L-1)}} a_u^{(L-1)} = \Delta_u^{(L-1)} \cdot \frac{\partial}{\partial w_{u,v}^{(L-1)}} w_{u,v}^{(L-1)} \cdot a_v^{(L-2)} = \Delta_u^{(L-1)} \cdot a_v^{(L-2)} \quad (18)$$

Plugging this back into equation 16 and then that into equation 14:

$$\frac{\partial}{\partial w_{u,v}^{(L-1)}} C(\mathbf{a}^{(0)}) = \sum_{j=1}^{n_L} (a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot w_{j,u}^{(L)} \cdot \Delta_u^{(L-1)} \cdot a_v^{(L-2)} \quad (19)$$

3.1.1.4 In-between hidden layers Partial derivatives for the k -th layer for $L \geq 3$ and $L-2 \geq k \in \mathbb{N}^+$ and $u, v \in \mathbb{N}, 1 \leq u \leq n_k, v \leq n_{k-1}$, using equation 5:

$$\begin{aligned} \frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) &= \frac{\partial}{\partial w_{u,v}^{(k)}} \sum_{j=1}^{n_L} \frac{1}{2} \cdot (a_j^{(L)} - y_j)^2 \\ &= \sum_{j=1}^{n_L} \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} (a_j^{(L)} - y_j)^2 \\ &= \sum_{j=1}^{n_L} \frac{1}{2} \cdot 2 \cdot (a_j^{(L)} - y_j) \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} a_j^{(L)} \\ &= \sum_{j=1}^{n_L} (a_j^{(L)} - y_j) \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} f_j^{(L)} \left(\sum_{i_1=0}^{n_{L-1}} w_{j,i_1}^{(L)} \cdot a_{i_1}^{(L-1)} \right) \\ &= \sum_{j=1}^{n_L} (a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot \left(\sum_{i_1=0}^{n_{L-1}} w_{j,i_1}^{(L)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_1}^{(L-1)} \right) \\ &= \sum_{j=1}^{n_L} (a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot \left(\sum_{i_1=0}^{n_{L-1}} w_{j,i_1}^{(L)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} f_{i_1}^{(L-1)} \left(\sum_{i_2=0}^{n_{L-2}} w_{i_1,i_2}^{(L-1)} \cdot a_{i_2}^{(L-2)} \right) \right) \\ &= \sum_{j=1}^{n_L} (a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot \left(\sum_{i_1=0}^{n_{L-1}} w_{j,i_1}^{(L)} \cdot \Delta_{i_1}^{(L-1)} \cdot \left(\sum_{i_2=0}^{n_{L-2}} w_{i_1,i_2}^{(L-1)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_2}^{(L-2)} \right) \right) \\ &= \sum_{j=1}^{n_L} \sum_{i_1=0}^{n_{L-1}} (a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot w_{j,i_1}^{(L)} \cdot \Delta_{i_1}^{(L-1)} \cdot \left(\sum_{i_2=0}^{n_{L-2}} w_{i_1,i_2}^{(L-1)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_2}^{(L-2)} \right) \\ &= \sum_{j=1}^{n_L} \sum_{i_1=0}^{n_{L-1}} \sum_{i_2=0}^{n_{L-2}} (a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot w_{j,i_1}^{(L)} \cdot \Delta_{i_1}^{(L-1)} \cdot w_{i_1,i_2}^{(L-1)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_2}^{(L-2)} \\ &= \dots \end{aligned} \quad (20)$$

Due to the chain rule and the commutativity of addition, the emerging pattern of introducing $\Delta_{i_{L-l}}^{(l)} \cdot w_{i_{L-l}, i_{L-l+1}}^{(l)}$ terms (where $l \in \mathbb{N}^+, k < l \leq L$) will continue as we go down to deeper and deeper layers, expanding the activations. Eventually, we reach the layer immediately above the k -th layer, where we need to find $\frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_{L-(k+1)}}^{(k+1)}$:

$$\begin{aligned}
\frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_{L-(k+1)}}^{(k+1)} &= \frac{\partial}{\partial w_{u,v}^{(k)}} f_{i_{L-(k+1)}}^{(k+1)} \left(\sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot a_{i_{L-k}}^{(k)} \right) \\
&= f_{i_{L-(k+1)}}^{(k+1)} \cdot \left(\sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot a_{i_{L-k}}^{(k)} \right) \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} \left(\sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot a_{i_{L-k}}^{(k)} \right) \\
&= \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot \sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_{L-k}}^{(k)} \\
&= \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot \sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} f_{i_{L-k}}^{(k)} \left(\sum_{i_{L-k+1}=0}^{n_{k-1}} w_{i_{L-k}, i_{L-k+1}}^{(k)} \cdot a_{i_{L-k+1}}^{(k-1)} \right) \\
&= \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot \sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot \Delta_{i_{L-k}}^{(k)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} \left(\sum_{i_{L-k+1}=0}^{n_{k-1}} w_{i_{L-k}, i_{L-k+1}}^{(k)} \cdot a_{i_{L-k+1}}^{(k-1)} \right) \\
&= \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot \sum_{i_{L-k}=0}^{n_k} w_{i_{L-(k+1)}, i_{L-k}}^{(k+1)} \cdot \Delta_{i_{L-k}}^{(k)} \cdot \left(\sum_{i_{L-k+1}=0}^{n_{k-1}} \frac{\partial}{\partial w_{u,v}^{(k)}} w_{i_{L-k}, i_{L-k+1}}^{(k)} \cdot a_{i_{L-k+1}}^{(k-1)} \right)
\end{aligned} \tag{21}$$

The only term which depends on $w_{u,v}^{(k)}$ in the inner summation is the one where $i_{L-k} = u$ and $i_{L-k+1} = v$, so the derivative of all the other terms is 0. This also makes the terms in the outer summation equal to 0 where $i_{L-k} \neq u$, so we're left with:

$$\begin{aligned}
\frac{\partial}{\partial w_{u,v}^{(k)}} a_{i_{L-(k+1)}}^{(k+1)} &= \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot w_{i_{L-(k+1)}, u}^{(k+1)} \cdot \Delta_u^{(k)} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} w_{u,v}^{(k)} \cdot a_v^{(k-1)} \\
&= \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot w_{i_{L-(k+1)}, u}^{(k+1)} \cdot \Delta_u^{(k)} \cdot a_v^{(k-1)}
\end{aligned} \tag{22}$$

Therefore the general form of equation 20 can be written as:

$$\begin{aligned}
\frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) &= \sum_{j=1}^{n_L} \sum_{i_1=0}^{n_{L-1}} \dots \sum_{i_{L-(k+1)}=0}^{n_{k+1}} \left(a_j^{(L)} - y_j \right) \\
&\quad \cdot \Delta_j^{(L)} \cdot w_{j, i_1}^{(L)} \\
&\quad \cdot \Delta_{i_1}^{(L-1)} \cdot w_{i_1, i_2}^{(L-1)} \\
&\quad \vdots \\
&\quad \cdot \Delta_{i_{L-(k+1)}}^{(k+1)} \cdot w_{i_{L-(k+1)}, u}^{(k+1)} \\
&\quad \cdot \Delta_u^{(k)} \cdot a_v^{(k-1)}
\end{aligned} \tag{23}$$

3.2 Basic backpropagation algorithm

As can be seen in equation 23, calculating the partial derivatives requires an exponentially growing number of additions and multiplications as more and more layers are added to a network. Fortunately, the values of the $(a_j^{(L)} - y_j) \cdot \Delta_j^{(L)} \cdot w_{j,i_1}^{(L)} \cdot \Delta_{i_1}^{(L-1)} \cdot w_{i_1,i_2}^{(L-1)} \cdot \dots \cdot \Delta_{i_{L-k}}^{(k)} \cdot w_{i_{L-k},i_{L-k+1}}^{(k)}$ products can be cached and reused for subsequent layers if we evaluate the activations across the network forward (from the input layer towards the output layer), and then calculate the gradient backwards (from the output layer towards the input layer). This backwards propagation of the error is the origin of the name of this learning algorithm.

In theory, we should evaluate all the examples and use the average of their gradients to adjust the network, then keep repeating this until we are satisfied with the network's performance. (Such an iteration is called an "epoch".) In practice, however, due to the huge amount of training data that may be needed to train a network, the examples are usually grouped in small batches randomly, and the network is adjusted for each batch instead of the whole set of examples. This way not all the steps of the gradient descent will be optimal, but the overall learning time can be a lot shorter.

Also, the step size is usually adjusted with a parameter $0 < \alpha \in \mathbb{R}$ called the "learning rate" or "braveness", which may vary over time. Initially the learning rate can be high (usually $0 < \alpha < 0.3$) so that the network makes bigger steps towards the optimum, and later it can be reduced gradually so that when the network is close to the optimum, the chance of overshooting can be minimized.

Another way to avoid overshooting may be to make the step size proportional to the length of the gradient vector.

Algorithm 1 to 10 show the basic backpropagation training algorithm procedures with both expanded and matrix-vector notation.

Algorithm 1 Train the network with the given set of examples. An example is an $(\mathbf{x}, \mathbf{y})$ pair where $\mathbf{x} \in \mathbb{R}^{n_0}$ is an example input, and $\mathbf{y} \in \mathbb{R}^{n_L}$ is the desired output for $\mathbf{x}$.

```

procedure LEARN(examples)
   $\alpha \leftarrow$  initial learning rate
  INITIALIZE
  repeat
    batch  $\leftarrow$  randomly chosen items from examples
    CLEARGRADIENT
    for  $(\mathbf{x}, \mathbf{y})$  in batch do
      EVALUATE( $\mathbf{x}$ )
      ADJUSTGRADIENT( $\mathbf{y}$ )
    end for
    APPLYGRADIENT( $\alpha$ )
     $\alpha \leftarrow$  new learning rate
  until training is done
end procedure

```

The stopping criteria can be that a fixed number of epochs have run, or that the error has become consistently acceptably low for most examples, or that the gradient (and thus, the step size) has become so close to zero that the algorithm has basically reached a flat region from where it will no longer move away any significantly.

Algorithm 2 Initialize the network with random weights and biases.

```
procedure INITIALIZE
  for  $k \leftarrow 0$  to  $L$  do
     $a_0^{(k)} \leftarrow 1$ 
     $d_0^{(k)} \leftarrow 1$ 
  end for
  for  $k \leftarrow 1$  to  $L$  do
    for  $i \leftarrow 0$  to  $n_{k-1}$  do
       $w_{0,i}^{(k)} \leftarrow 1$ 
      for  $j \leftarrow 1$  to  $n_k$  do
         $w_{j,i}^{(k)} \leftarrow \text{random number}$ 
         $\nabla C_{j,i}^{(k)} \leftarrow 0$ 
      end for
    end for
  end for
end procedure
```

Algorithm 3 Initialize the gradient that belongs to the current batch.

```
procedure CLEARGRADIENT
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $\nabla C_{j,i}^{(k)} \leftarrow 0$ 
      end for
    end for
  end for
end procedure
```

Algorithm 4 Algorithm 4 with matrix-vector notation.

```
procedure CLEARGRADIENT
  for  $k \leftarrow 1$  to  $L$  do
     $\nabla \mathbf{C}^{(k)} \leftarrow \mathbf{0}$ 
  end for
end procedure
```

Algorithm 5 Evaluate the given input and produce the result in the output layer.

```
procedure EVALUATE( $\mathbf{x}$ )
  for  $j \leftarrow 1$  to  $n_0$  do
     $a_j^{(0)} \leftarrow x_j$ 
  end for
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
       $s_j^{(k)} \leftarrow \sum_{i=0}^{n_{k-1}} w_{j,i}^{(k)} \cdot a_i^{(k-1)}$ 
       $a_j^{(k)} \leftarrow f_j^{(k)}(s_j^{(k)})$ 
    end for
  end for
end procedure
```

Algorithm 6 Algorithm 5 with matrix-vector notation.

```

procedure EVALUATE( $\mathbf{x}$ )
   $\mathbf{a}^{(0)} \leftarrow \mathbf{x}$ 
  for  $k \leftarrow 1$  to  $L$  do
     $\mathbf{s}^{(k)} \leftarrow \mathbf{W}^{(k)} \mathbf{a}^{(k-1)}$ 
     $\mathbf{a}^{(k)} \leftarrow \mathbf{f}^{(k)}(\mathbf{s}^{(k)})$ 
  end for
end procedure

```

Algorithm 7 Calculate the gradient with error backpropagation for the given expected output ($\mathbf{y} \in \mathbb{R}^{n_L}$), and merge it into the gradient of the current batch.

```

procedure ADJUSTGRADIENT( $\mathbf{y}$ )
  for  $j \leftarrow 1$  to  $n_L$  do
     $d_j^{(L)} \leftarrow (y_j - a_j^{(L)}) \cdot f_j^{(L)'}(s_j^{(L)})$ 
    for  $i \leftarrow 0$  to  $n_{L-1}$  do
       $\nabla C_{j,i}^{(L)} \leftarrow \nabla C_{j,i}^{(L)} + d_j^{(L)} \cdot a_i^{(L-1)}$ 
    end for
  end for
  for  $k \leftarrow L-1$  to  $1$  do
    for  $j \leftarrow 1$  to  $n_k$  do
       $d_j^{(k)} \leftarrow \left( \sum_{i=0}^{n_{k+1}} d_i^{(k+1)} \cdot w_{i,j}^{(k+1)} \right) \cdot f_j^{(k)'}(s_j^{(k)})$ 
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $\nabla C_{j,i}^{(k)} \leftarrow \nabla C_{j,i}^{(k)} + d_j^{(k)} \cdot a_i^{(k-1)}$ 
      end for
    end for
  end for
end procedure

```

Algorithm 8 Algorithm 7 with matrix-vector notation.

```

procedure ADJUSTGRADIENT( $\mathbf{y}$ )
   $\mathbf{d}^{(L)} \leftarrow (\mathbf{y} - \mathbf{a}^{(L)}) \odot \mathbf{f}^{(L)'}(\mathbf{s}^{(L)})$ 
   $\nabla \mathbf{C}^{(L)} \leftarrow \nabla \mathbf{C}^{(L)} + \mathbf{d}^{(L)} (\mathbf{a}^{(L-1)})^T$  ▷ matrix multiplication
  for  $k \leftarrow L-1$  to  $1$  do
     $\mathbf{d}^{(k)} \leftarrow \left( (\mathbf{W}^{(k+1)})^T \mathbf{d}^{(k+1)} \right) \odot \mathbf{f}^{(k)'}(\mathbf{s}^{(k)})$ 
     $\nabla \mathbf{C}^{(k)} \leftarrow \nabla \mathbf{C}^{(k)} + \mathbf{d}^{(k)} (\mathbf{a}^{(k-1)})^T$  ▷ matrix multiplication
  end for
end procedure

```

Algorithm 9 Adjust the weights and biases of the network with the gradient of the current batch and the given learning rate.

```

procedure APPLYGRADIENT( $\alpha$ )
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $w_{j,i}^{(k)} \leftarrow w_{j,i}^{(k)} - \alpha \cdot \nabla C_{j,i}^{(k)}$ 
      end for
    end for
  end for
end procedure

```

Algorithm 10 Algorithm 9 with matrix-vector notation.

```

procedure APPLYGRADIENT( $\alpha$ )
  for  $k \leftarrow 1$  to  $L$  do
     $\mathbf{W}^{(k)} \leftarrow \mathbf{W}^{(k)} - \alpha \cdot \nabla \mathbf{C}^{(k)}$ 
  end for
end procedure

```

3.3 Training and testing (validating)

The goal is usually not to have the network memorize all the training examples and then spit out memorized outputs (known as "overfitting"), but to produce acceptable output for unseen inputs as well.

To measure how well the network performs for unseen inputs, and to see if any adjustments are needed for the training process or the network's architecture, the examples with the known outputs are usually split into two subsets: the training data and the validation data (also known as test data). Usually, the training data contains more elements than the validation data, e.g. with a 2 : 1 or a 3 : 1 ratio.

The learning algorithm is run exclusively for the training examples, and then the error is measured against the validation examples, without making any further adjustments on the weights and biases.

Algorithm 11 is a modification of algorithm 1 in which the error for the training set is accumulated in E_t for each iteration, and the error for the validation set is accumulated in E_v .

3.3.1 Underfitting

When both E_t and E_v remains high, that can indicate a problem called "underfitting". This can occur when the network is too simple:

- it has too few input neurons,
- or it has too few neurons in too few layers,
- or too many weights and biases are close to zero (ie. too few neurons and features are actually used, there are too few connections in the network),
- or just too few epochs have been run.

3.3.2 Overfitting, regularization

A symptom of overfitting is when E_t is low, but E_v is significantly higher than E_t , ie. when the network performs well on the training set, but performs poorly on the validation set. It can indicate

Algorithm 11 Modified version of algorithm 1 with validation.

```
procedure LEARN(examples)
   $\alpha \leftarrow$  initial learning rate
  INITIALIZE
  trainers  $\leftarrow$  randomly chosen items from examples
  validators  $\leftarrow$  items from examples that are not in trainers
  repeat
    trainer batch  $\leftarrow$  randomly chosen items from trainers
    validator batch  $\leftarrow$  randomly chosen items from validators
    CLEARGRADIENT
    CALCULATEGRADIENT(trainer batch)
    APPLYGRADIENT( $\alpha$ )
    VALIDATE(validator batch)
     $\alpha \leftarrow$  new learning rate
  until training is done
end procedure
procedure CALCULATEGRADIENT(batch)
   $E_t \leftarrow 0$ 
  for (x, y) in batch do
    EVALUATE(x)
    ADJUSTGRADIENT(y)
     $E_t \leftarrow E_t + \frac{1}{2} \cdot \|\mathbf{y} - \mathbf{a}^{(L)}\|_2^2$ 
  end for
end procedure
procedure VALIDATE(batch)
   $E_v \leftarrow 0$ 
  for (x, y) in batch do
    EVALUATE(x)
     $E_v \leftarrow E_v + \frac{1}{2} \cdot \|\mathbf{y} - \mathbf{a}^{(L)}\|_2^2$ 
  end for
end procedure
```

that the network is too complex. Mitigation techniques are called "regularization". They may be applied only for the input layer, so the network will ignore some features and add more emphasis on others, or to the entire network.

3.3.2.1 Norm-based regularization The idea of this technique is to put all the weights and biases from all layers into a giant vector, and include its norm in the cost function with some weight.

Formally: let N denote the number of all weights and biases in the network. Let $\mathbf{w} \in \mathbb{R}^N$ be a vector which contains all $w_{i,j}^{(k)}$ weights and biases in the network for $i, j, k \in \mathbb{N}, 1 \leq k \leq L, 1 \leq i \leq n_k, j \leq n_{k-1}$. Let $\lambda \in \mathbb{R}$ be the regularization parameter. Then the regularized cost function with some vector norm:

$$C_r(\mathbf{a}^{(0)}) = C(\mathbf{a}^{(0)}) + \lambda \cdot \|\mathbf{w}\| \quad (24)$$

Algorithm 12 incorporates norm based regularization into the calculation of the gradient.

Algorithm 12 Modified version of algorithm 7 with norm-based regularization.

```

procedure ADJUSTGRADIENT( $\mathbf{y}$ )
  for  $j \leftarrow 1$  to  $n_L$  do
     $d_j^{(L)} \leftarrow (y_j - a_j^{(L)}) \cdot f_j^{(L)'}(s_j^{(L)})$ 
    for  $i \leftarrow 0$  to  $n_{L-1}$  do
       $R \leftarrow \lambda \cdot \text{REGULARIZATION}(L, j, i)$ 
       $\nabla C_{j,i}^{(L)} \leftarrow \nabla C_{j,i}^{(L)} + d_j^{(L)} \cdot a_i^{(L-1)} + R$ 
    end for
  end for
  for  $k \leftarrow L-1$  to  $1$  do
    for  $j \leftarrow 1$  to  $n_k$  do
       $d_j^{(k)} \leftarrow \left( \sum_{i=0}^{n_{k+1}} d_i^{(k+1)} \cdot w_{i,j}^{(k+1)} \right) \cdot f_j^{(k)'}(s_j^{(k)})$ 
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $R \leftarrow \lambda \cdot \text{REGULARIZATION}(k, j, i)$ 
         $\nabla C_{j,i}^{(k)} \leftarrow \nabla C_{j,i}^{(k)} + d_j^{(k)} \cdot a_i^{(k-1)} + R$ 
      end for
    end for
  end for
end procedure

```

3.3.2.1.1 L_1 regularization For $L \geq k \in \mathbb{N}^+$ and $u, v \in \mathbb{N}, 1 \leq u \leq n_k, v \leq n_{k-1}$:

$$\begin{aligned}
\frac{\partial}{\partial w_{u,v}^{(k)}} C_{L_1}(\mathbf{a}^{(0)}) &= \frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) + \frac{\partial}{\partial w_{u,v}^{(k)}} \lambda \cdot \|\mathbf{w}\|_1 \\
&= \frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) + \frac{\partial}{\partial w_{u,v}^{(k)}} \lambda \cdot \sum_{l=1}^L \sum_{j=1}^{n_l} \sum_{i=0}^{n_{l-1}} |w_{j,i}^{(l)}| \\
&= \frac{\partial}{\partial w_{u,v}^{(l)}} C(\mathbf{a}^{(0)}) + \lambda \cdot \sum_{l=1}^L \sum_{j=1}^{n_l} \sum_{i=0}^{n_{l-1}} \frac{\partial}{\partial w_{u,v}^{(k)}} |w_{j,i}^{(l)}|
\end{aligned} \quad (25)$$

Most of the $w_{j,i}^{(l)}$ terms don't depend on $w_{u,v}^{(k)}$, therefore their derivative is 0, and the only one which remains is the one which contains $w_{u,v}^{(k)}$:

$$\frac{\partial}{\partial w_{u,v}^{(k)}} C_{L_1}(\mathbf{a}^{(0)}) = \frac{\partial}{\partial w_{u,v}^{(l)}} C(\mathbf{a}^{(0)}) + \lambda \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} |w_{u,v}^{(k)}| \quad (26)$$

Since a connection with zero weight does not contribute to the network's complexity, we can interpret $|0|'$ as 0:

$$|x|' = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} \quad (27)$$

In other words, we modify algorithm 7 so that for each $C_{j,i}^{(k)}$ value of the gradient, we add or subtract λ , depending on whether $w_{j,i}^{(k)}$ is positive or negative.

Algorithm 13 implements L_1 regularization for algorithm 12.

Algorithm 13 L_1 regularization

```

function REGULARIZATION( $k, j, i$ )
  if  $w_{j,i}^{(k)} = 0$  then
    return 0
  end if
  if  $w_{j,i}^{(k)} > 0$  then
    return 1
  end if
  return -1
end function

```

3.3.2.1.2 L_2 regularization To make calculating the partial derivatives cleaner, we apply a slight modification to the regularization term: for $L \geq k \in \mathbb{N}^+$ and $u, v \in \mathbb{N}, 1 \leq u \leq n_k, v \leq n_{k-1}$:

$$\begin{aligned}
\frac{\partial}{\partial w_{u,v}^{(k)}} C_{L_1}(\mathbf{a}^{(0)}) &= \frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) + \frac{\partial}{\partial w_{u,v}^{(k)}} \lambda \cdot \frac{1}{2} \cdot \|\mathbf{w}\|_2^2 \\
&= \frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) + \frac{\partial}{\partial w_{u,v}^{(k)}} \lambda \cdot \frac{1}{2} \cdot \left(\sqrt{\sum_{l=1}^L \sum_{j=1}^{n_l} \sum_{i=0}^{n_{l-1}} (w_{j,i}^{(l)})^2} \right)^2 \\
&= \frac{\partial}{\partial w_{u,v}^{(k)}} C(\mathbf{a}^{(0)}) + \frac{\partial}{\partial w_{u,v}^{(k)}} \lambda \cdot \frac{1}{2} \cdot \sum_{l=1}^L \sum_{j=1}^{n_l} \sum_{i=0}^{n_{l-1}} (w_{j,i}^{(l)})^2 \\
&= \frac{\partial}{\partial w_{u,v}^{(l)}} C(\mathbf{a}^{(0)}) + \lambda \cdot \frac{1}{2} \cdot \sum_{l=1}^L \sum_{j=1}^{n_l} \sum_{i=0}^{n_{l-1}} \frac{\partial}{\partial w_{u,v}^{(k)}} (w_{j,i}^{(l)})^2
\end{aligned} \quad (28)$$

Again, most of the $w_{j,i}^{(l)}$ terms don't depend on $w_{u,v}^{(k)}$, therefore their derivative is 0, and the only one which remains is the one which contains $w_{u,v}^{(k)}$:

$$\begin{aligned}
\frac{\partial}{\partial w_{u,v}^{(k)}} C_{L_1}(\mathbf{a}^{(0)}) &= \frac{\partial}{\partial w_{u,v}^{(l)}} C(\mathbf{a}^{(0)}) + \lambda \cdot \frac{1}{2} \cdot \frac{\partial}{\partial w_{u,v}^{(k)}} \left(w_{u,v}^{(k)} \right)^2 \\
&= \frac{\partial}{\partial w_{u,v}^{(l)}} C(\mathbf{a}^{(0)}) + \lambda \cdot \frac{1}{2} \cdot 2 \cdot w_{u,v}^{(k)} \\
&= \frac{\partial}{\partial w_{u,v}^{(l)}} C(\mathbf{a}^{(0)}) + \lambda \cdot w_{u,v}^{(k)}
\end{aligned} \tag{29}$$

Algorithm 14 implements L_2 regularization for algorithm 12.

Algorithm 14 L_2 regularization

```

function REGULARIZATION( $k, j, i$ )
  return  $w_{j,i}^{(k)}$ 
end function

```

3.3.2.2 Dropout regularization Overfitting might be the result of some neurons making mistakes that are covered up by the others, so neurons start to develop dependence on each other's mistakes and corrections. This is called "complex co-adaptation".

To make sure that neurons don't depend on each other too much, this technique picks a certain number of neurons in each layer for each training example, and nullifies their contributions to the network, while making up for it by increasing the contribution of the remaining neurons.

A new evaluation procedure has to be introduced in algorithm 11, to be used only in the training step, but not during validation.

With $\delta \in \mathbb{R}, 0 \leq \delta < 1$ denoting the ratio of neurons in each layer that are dropped, algorithm 15 to 19 extend algorithm 11 with dropout regularization (shown with both expanded and matrix-vector notation).

Algorithm 15 Modified version of algorithm 11 with dropout regularization.

```

procedure CALCULATEGRADIENT( $batch$ )
   $E_t \leftarrow 0$ 
  for ( $\mathbf{x}, \mathbf{y}$ ) in  $batch$  do
    EVALUATEWITHDROPOUT( $\mathbf{x}$ )
    ADJUSTGRADIENTWITHDROPOUT( $\mathbf{y}$ )
     $E_t \leftarrow E_t + \frac{1}{2} \cdot \|\mathbf{y} - \mathbf{a}^{(L)}\|_2^2$ 
  end for
end procedure

```

Algorithm 16 Modified version of algorithm 5 with dropout regularization.

```

procedure EVALUATEWITHDROPOUT( $\mathbf{x}$ )
   $r_0^{(0)} \leftarrow 1$ 
  for  $j \leftarrow 1$  to  $n_0$  do
     $r_j^{(0)} \leftarrow 0$  with  $\delta$  chance ,  $\frac{1}{1-\delta}$  with  $1 - \delta$  chance
     $a_j^{(0)} \leftarrow r_j^{(0)} \cdot x_j$ 
  end for
  for  $k \leftarrow 1$  to  $L$  do
     $r_0^{(k)} \leftarrow 1$ 
    for  $j \leftarrow 1$  to  $n_k$  do
       $r_j^{(k)} \leftarrow 0$  with  $\delta$  chance ,  $\frac{1}{1-\delta}$  with  $1 - \delta$  chance
      if  $k = L$  or  $r_j^{(k)} > 0$  then
         $s_j^{(k)} \leftarrow \sum_{i=0}^{n_{k-1}} w_{j,i}^{(k)} \cdot a_i^{(k-1)}$ 
         $a_j^{(k)} \leftarrow r_j^{(k)} \cdot f_j^{(k)}(s_j^{(k)})$ 
      else
         $s_j^{(k)} \leftarrow 0$ 
         $a_j^{(k)} \leftarrow 0$ 
      end if
    end for
  end for
end procedure

```

Algorithm 17 Algorithm 16 with matrix-vector notation.

```

procedure EVALUATEWITHDROPOUT( $\mathbf{x}$ )
   $\mathbf{r}^{(0)} \leftarrow \left[ 0 \text{ with } \delta \text{ chance , } \frac{1}{1-\delta} \text{ with } 1 - \delta \text{ chance} \right]_{i=1}^{n_0}$ 
   $r_0^{(0)} \leftarrow 1$ 
   $\mathbf{a}^{(0)} \leftarrow \mathbf{r} \odot \mathbf{x}$ 
  for  $k \leftarrow 1$  to  $L$  do
    if  $k = L$  then
       $\mathbf{s}^{(k)} \leftarrow \mathbf{W}^{(k)} \mathbf{a}^{(k-1)}$ 
       $\mathbf{a}^{(k)} \leftarrow \mathbf{f}^{(k)}(\mathbf{s}^{(k)})$ 
    else
       $\mathbf{r}^{(k)} \leftarrow \left[ 0 \text{ with } \delta \text{ chance , } \frac{1}{1-\delta} \text{ with } 1 - \delta \text{ chance} \right]_{i=1}^{n_k}$ 
       $r_0^{(k)} \leftarrow 1$ 
       $\mathbf{s}^{(k)} \leftarrow \mathbf{r}^{(k)} \odot (\mathbf{W}^{(k)} \mathbf{a}^{(k-1)})$ 
       $\mathbf{a}^{(k)} \leftarrow \mathbf{r}^{(k)} \odot \mathbf{f}^{(k)}(\mathbf{s}^{(k)})$ 
    end if
  end for
end procedure

```

Algorithm 18 Modified version of algorithm 7 with dropout regularization.

```

procedure ADJUSTGRADIENTWITHDROPOUT(y)
  for  $j \leftarrow 1$  to  $n_L$  do
     $d_j^{(L)} \leftarrow (y_j - a_j^{(L)}) \cdot f_j^{(L)'}(s_j^{(L)})$ 
    for  $i \leftarrow 0$  to  $n_{L-1}$  do
       $\nabla C_{j,i}^{(L)} \leftarrow \nabla C_{j,i}^{(L)} + d_j^{(L)} \cdot a_i^{(L-1)}$ 
    end for
  end for
  for  $k \leftarrow L - 1$  to 1 do
    for  $j \leftarrow 1$  to  $n_k$  do
      if  $r_j^{(k)} > 0$  then
         $d_j^{(k)} \leftarrow r_j^{(k)} \cdot \left( \sum_{i=0}^{n_{k+1}} d_i^{(k+1)} \cdot w_{i,j}^{(k+1)} \right) \cdot f_j^{(k)'}(s_j^{(k)})$ 
        for  $i \leftarrow 0$  to  $n_{k-1}$  do
           $\nabla C_{j,i}^{(k)} \leftarrow \nabla C_{j,i}^{(k)} + d_j^{(k)} \cdot a_i^{(k-1)}$ 
        end for
      else
         $d_j^{(k)} \leftarrow 0$ 
      end if
    end for
  end for
end procedure

```

Algorithm 19 Algorithm 18 with matrix-vector notation.

```

procedure ADJUSTGRADIENTWITHDROPOUT(y)
   $\mathbf{d}^{(L)} \leftarrow (\mathbf{y} - \mathbf{a}^{(L)}) \odot \mathbf{f}^{(L)'}(\mathbf{s}^{(L)})$ 
   $\nabla \mathbf{C}^{(L)} \leftarrow \nabla \mathbf{C}^{(L)} + \mathbf{d}^{(L)} (\mathbf{a}^{(L-1)})^T$  ▷ matrix multiplication
  for  $k \leftarrow L - 1$  to 1 do
     $\mathbf{d}^{(k)} \leftarrow \mathbf{r}^{(k)} \odot \left( (\mathbf{W}^{(k+1)})^T \mathbf{d}^{(k+1)} \right) \odot \mathbf{f}^{(k)'}(\mathbf{s}^{(k)})$ 
     $\nabla \mathbf{C}^{(k)} \leftarrow \nabla \mathbf{C}^{(k)} + \mathbf{d}^{(k)} (\mathbf{a}^{(k-1)})^T$  ▷ matrix multiplication
  end for
end procedure

```

3.4 Optimizers

3.4.1 Momentum (inertia)

This technique can sometimes speed up the learning process, and help with preventing the algorithm getting stuck in local minima. The idea is to smoothen the direction changes of the gradient descent algorithm, so that when a new gradient is calculated for a new training batch, the algorithm keeps moving towards the previous gradient to some degree as well. One way to achieve this is to store the gradient vector before clearing it for the new batch, and calculate a weighted average of the previous and the new gradient in the ApplyGradient procedure (algorithm 9, 10) before modifying the weights.

With $\beta \in \mathbb{R}, 0 \leq \beta < 1$ denoting the amount to keep from the previous gradient (usually chosen to be close to 1, e.g. 0.8, 0.9, or even 0.99), algorithm 20 is a modification of algorithm 4 and 9 with momentum.

Algorithm 20 Modified version of algorithm 4 and 9 with momentum.

```

procedure CLEARGRADIENT
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $\nabla C_{j,i}^{(k)\text{prev}} \leftarrow \nabla C_{j,i}^{(k)}$ 
         $\nabla C_{j,i}^{(k)} \leftarrow 0$ 
      end for
    end for
  end for
end procedure

procedure APPLYGRADIENT( $\alpha$ )
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $\nabla C_{j,i}^{(k)} \leftarrow \beta \cdot \nabla C_{j,i}^{(k)\text{prev}} + (1 - \beta) \cdot \nabla C_{j,i}^{(k)}$ 
         $w_{j,i}^{(k)} \leftarrow w_{j,i}^{(k)} - \alpha \cdot \nabla C_{j,i}^{(k)}$ 
      end for
    end for
  end for
end procedure

```

Algorithm 21 Algorithm 20 with matrix-vector notation.

```

procedure CLEARGRADIENT
  for  $k \leftarrow 1$  to  $L$  do
     $\nabla \mathbf{C}^{(k)\text{prev}} \leftarrow \nabla \mathbf{C}^{(k)}$ 
     $\nabla \mathbf{C}^{(k)} \leftarrow \mathbf{0}$ 
  end for
end procedure

procedure APPLYGRADIENT( $\alpha, \beta$ )
  for  $k \leftarrow 1$  to  $L$  do
     $\nabla \mathbf{C}^{(k)} \leftarrow \beta \cdot \nabla \mathbf{C}^{(k)\text{prev}} + (1 - \beta) \cdot \nabla \mathbf{C}^{(k)}$ 
     $\mathbf{W}^{(k)} \leftarrow \mathbf{W}^{(k)} - \alpha \cdot \nabla \mathbf{C}^{(k)}$ 
  end for
end procedure

```

3.4.2 RMSProp (Root Mean Squared Propagation)

Similarly to the Momentum optimizer (3.4.1), the goal of RMSProp is to smoothen the zig-zag path of the mini-batch gradient descent. The idea is to adjust the learning rate based on the current and the previous values of the gradient: small weight adjustments will be scaled up in order to speed up convergence, large weight adjustments will be decreased in order to prevent overshooting.

Algorithm 22 is the modification of algorithm 2 and 9 with RMSProp. The $\beta \in \mathbb{R}, 0 \leq \beta < 1$ parameter is called the "decay rate", and it is usually set to be around 0.9 or 0.999. The purpose of the $0 < \varepsilon \in \mathbb{R}$ parameter is to avoid division by zero, and to prevent floating point rounding errors from making the computation numerically unstable; its value is usually chosen to be around 10^{-6} or 10^{-8} .

Algorithm 22 Modified version of algorithm 2 and 9 with RMSProp.

```

procedure INITIALIZE
  for  $k \leftarrow 0$  to  $L$  do
     $a_0^{(k)} \leftarrow 1$ 
     $d_0^{(k)} \leftarrow 1$ 
  end for
  for  $k \leftarrow 1$  to  $L$  do
    for  $i \leftarrow 0$  to  $n_{k-1}$  do
       $w_{0,i}^{(k)} \leftarrow 1$ 
      for  $j \leftarrow 1$  to  $n_k$  do
         $w_{j,i}^{(k)} \leftarrow \text{random number}$ 
         $S_{j,i}^{(k)} \leftarrow 0$ 
         $\nabla C_{j,i}^{(k)} \leftarrow 0$ 
      end for
    end for
  end for
end procedure

procedure APPLYGRADIENT( $\alpha, \beta, \varepsilon$ )
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $S_{j,i}^{(k)} \leftarrow \beta \cdot S_{j,i}^{(k)} + (1 - \beta) \cdot \left( \nabla C_{j,i}^{(k)} \right)^2$ 
         $w_{j,i}^{(k)} \leftarrow w_{j,i}^{(k)} - \frac{\alpha}{\sqrt{S_{j,i}^{(k)} + \varepsilon}} \cdot \nabla C_{j,i}^{(k)}$ 
      end for
    end for
  end for
end procedure

```

3.4.3 Adam (ADaptive Moment estimation)

The Adam optimizer combines the advantages of the Momentum optimizer (3.4.1) and the RMSProp optimizer (3.4.2) by doing both at the same time.

Algorithm 23 shows a variation of algorithm 9 with Adam optimizer. The Initialize procedure is the same as in algorithm 22, and the ClearGradient procedure is the same as in 20.

The $0 < \varepsilon \in \mathbb{R}$ parameter is the same as in algorithm 22, and the $\beta_1, \beta_2 \in \mathbb{R}, 0 \leq \beta_1, \beta_2 < 1$ parameters are chosen similarly to the ones in the case of the Momentum optimizer (3.4.1) and the RMSProp optimizer (3.4.2) respectively.

Algorithm 23 Modified version of algorithm 9 with Adam optimizer.

```

procedure APPLYGRADIENT( $\alpha, \beta_1, \beta_2, \varepsilon$ )
  for  $k \leftarrow 1$  to  $L$  do
    for  $j \leftarrow 1$  to  $n_k$  do
      for  $i \leftarrow 0$  to  $n_{k-1}$  do
         $S_{j,i}^{(k)} \leftarrow \beta_2 \cdot S_{j,i}^{(k)} + (1 - \beta_2) \cdot \left( \nabla C_{j,i}^{(k)} \right)^2$  ▷ RMSProp
         $\nabla C_{j,i}^{(k)} \leftarrow \beta_1 \cdot \nabla C_{j,i}^{(k)\text{prev}} + (1 - \beta_1) \cdot \nabla C_{j,i}^{(k)}$  ▷ Momentum
         $w_{j,i}^{(k)} \leftarrow w_{j,i}^{(k)} - \frac{\alpha}{\sqrt{S_{j,i}^{(k)}} + \varepsilon} \cdot \nabla C_{j,i}^{(k)}$ 
      end for
    end for
  end for
end procedure

```

4 Advanced topics

4.1 Preprocessing

Depending on several factors (e.g. the problem at hand, the choice of activation functions, the format and the interpretation of the input and output data, numerical considerations, etc.), various kinds of preprocessing can significantly improve the network's performance.

4.1.1 Statistical methods

For $x \in \mathbb{R}$, when x is sufficiently far from 0, most of the activation functions (2.1.1) behave linearly in the sense that for $0 < c \in \mathbb{R}$, $f(c \cdot x) = c \cdot f(x)$. This implies that if the entire input of the network is scaled by a factor of c , then its output will also be scaled by the same amount. If the expected output remains fixed, then this scaling effect will be compensated by the training process by introducing a scaling factor of $\frac{1}{c}$ into the weights and biases in the network. It will also be reflected in the gradient, thus, it will affect the optimal choice of the learning rate as well. If the network has to do too much of this compensation, then it will be more susceptible to issues like the "vanishing gradient problem" where the gradient becomes so small for most inputs that basically no learning occurs. This, together with floating point arithmetics considerations in practical implementations, suggests that in most cases, it can be beneficial to bring the network's input closer to the $[0, 1]$ or the $[-1, 1]$ intervals.

4.1.1.1 Normalization Let $u, l \in \mathbb{R}$ (typically, $l = 1$ and $u \in \{-1, 0\}$), and let:

$$\begin{aligned}
 x_{\min} &= \min \{x_i : (\mathbf{x}, \mathbf{y}) \in \text{trainers}, i \in \{1, 2, \dots, n_0\}\} \\
 x_{\max} &= \max \{x_i : (\mathbf{x}, \mathbf{y}) \in \text{trainers}, i \in \{1, 2, \dots, n_0\}\}
 \end{aligned}$$

Note that only the *trainers* set is used from algorithm 11 for determining these values, since we don't want any information to be leaked into the training process from the validation set.

Before the network is evaluated for a given $\mathbf{x}$ input, it is first transformed with the following formula (for $j \in \{1, 2, \dots, n_0\}$):

$$\hat{x}_j = \frac{x_j - x_{\min}}{x_{\max} - x_{\min}} \cdot (u - l) + l \text{ for } j \in \{1, 2, \dots, n_0\}$$

Note: if the input is composed from heterogeneous features from different ranges, then it might be a good idea to normalize them separately or by feature groups. For $j \in \{1, 2, \dots, n_0\}$ and $u_j, l_j \in \mathbb{R}$, let

$$x_{j \min} = \min \{x_j : (\mathbf{x}, \mathbf{y}) \in \text{trainers}\}$$

$$x_{j \max} = \max \{x_j : (\mathbf{x}, \mathbf{y}) \in \text{trainers}\}$$

Then

$$\hat{x}_j = \frac{x_j - x_{j \min}}{x_{j \max} - x_{j \min}} \cdot (u_j - l_j) + l_j$$

4.1.1.2 Standardization For the analysis of statistical data, the exact value of its samples, its mean, and its deviation from the mean are often less important than the shape of its distribution. Standardization is a technique which brings statistical data into a single well defined location on the number line so that its distribution can be analyzed in well defined circumstances.

Standardization requires the mean and the standard deviation of the theoretical distribution to be known, but this is rarely the case in practice, so their estimated values are used instead. Let $N \in \mathbb{N}^+$ denote the number of examples in the *training* set:

$$\mu = \frac{1}{N \cdot n_0} \cdot \sum_{\mathbf{x} \in \text{training}} \sum_{i=0}^{n_0} x_i$$

$$\sigma = \sqrt{\frac{1}{N \cdot n_0 - 1} \cdot \sum_{\mathbf{x} \in \text{training}} \sum_{i=0}^{n_0} (x_i - \mu)^2}$$

Then, before the network is evaluated for a given $\mathbf{x}$ input, it is first transformed with the following formula (for $j \in \{1, 2, \dots, n_0\}$):

$$\hat{x}_j = \frac{x_j - \mu}{\sigma}$$

Again, if the input is composed of heterogeneous features with different ranges and distributions, then standardization is also better performed separately for the different features (or feature groups). For $j \in \{1, 2, \dots, n_0\}$:

$$\mu_j = \frac{1}{N} \cdot \sum_{\mathbf{x} \in \text{training}} x_j$$

$$\sigma_j = \sqrt{\frac{1}{N - 1} \cdot \sum_{\mathbf{x} \in \text{training}} (x_j - \mu_j)^2}$$

$$\hat{x}_j = \frac{x_j - \mu_j}{\sigma_j}$$

4.1.1.3 Batch normalization TODO

4.2 Convolutional Neural Networks

TODO

4.3 Recurrent Neural Networks

TODO

A Prerequisites and notation

Here's a quick recap of the calculus and linear algebra that is necessary for neural networks. For more precise definitions and the proofs of the theorems that are stated here without one, refer to any calculus and linear algebra textbook.

A.1 The set of natural numbers: $\mathbb{N}, \mathbb{N}^+$

$\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{N}^+ = \{1, 2, \dots\}$

A.2 The set of integer numbers: $\mathbb{Z}$

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

A.3 The set of real numbers: $\mathbb{R}$

Denoted by $\mathbb{R}$. For a precise definition, refer to any calculus textbook. The gist of it is that $\mathbb{R}$ is the number line, and it doesn't have any gaps and discontinuities.

A.4 Element in a set: $x \in \mathbb{R}$

$x \in \mathbb{R}$ means x is a real number.

A.5 Subset: $A \subset B$

If A and B are sets, then A is a subset of B if and only if for all $x \in A$ it holds that $x \in B$. Notation: $A \subset B$.

A.6 Intersection: $A \cap B$

If A and B are sets, then the intersection of A and B is a set which contains all elements of A that are also an element of B . Formally:

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

A.7 Interval: $[a, b], (a, b), [a, b), (a, b]$

A subset of the $\mathbb{R}$ that contains all real numbers lying between its two endpoints. For $a, b \in \mathbb{R}, a \leq b$:

- Open interval:

$$(a, b) = \{x : x \in \mathbb{R} \text{ and } a < x < b\}$$

If $a = b$, then (a, b) is the empty set.

- Closed interval:

$$[a, b] = \{x : x \in \mathbb{R} \text{ and } a \leq x \leq b\}$$

If $a = b$, then $[a, b]$ contains only a single number, a .

- Half-open intervals:

$$(a, b] = \{x : x \in \mathbb{R} \text{ and } a < x \leq b\}$$

$$[a, b) = \{x : x \in \mathbb{R} \text{ and } a \leq x < b\}$$

- Special intervals:

$$(-\infty, b) = \{x : x \in \mathbb{R} \text{ and } x < b\}$$

$$(a, +\infty) = \{x : x \in \mathbb{R} \text{ and } a < x\}$$

$$(-\infty, b] = \{x : x \in \mathbb{R} \text{ and } x \leq b\}$$

$$[a, +\infty) = \{x : x \in \mathbb{R} \text{ and } a \leq x\}$$

$$(-\infty, +\infty) = \mathbb{R}$$

A.8 Cartesian product of sets: $A \times B$, A^n

If A and B are sets, then $A \times B$ is a set which contains all possible pairings of an element from A and another one from B (in that order):

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

This can be extended to any number of sets, to produce n -tuples of any length. For a given $n \in \mathbb{N}$, $1 < n$, let $A_1, A_2, \dots, A_n$ be sets. Then:

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) : a_1 \in A_1 \text{ and } a_2 \in A_2 \text{ and } \dots \text{ and } a_n \in A_n\}$$

Specifically, if $A = A_1 = A_2 = \dots = A_n$ in the above example, then A^n is a shorthand notation for $A_1 \times A_2 \times \dots \times A_n$, ie. the set of all n -tuples that can be constructed from the elements of A .

A.9 Vector: $\mathbf{x} \in \mathbb{R}^n$

For $n \in \mathbb{N}^+$, $\mathbf{a} \in \mathbb{R}^n$ is an n -dimensional vector that contains the $a_i \in \mathbb{R}$ numbers (for $n \geq i \in \mathbb{N}^+$), ie. a list of numbers:

$$\mathbf{a} = [a_i]_{i=1}^n = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

Notes:

- A vector can be thought of as a point in an n -dimensional space.
- An n -tuple of real numbers and an n -dimensional real vector are basically the same thing; n -tuples focus more on set theory, while vectors put more emphasis on the geometrical, linear algebra interpretation.

- $\mathbf{0} \in \mathbb{R}^n$ denotes the vector in which all components are zero (called a "null vector" or a "zero vector"):

$$\mathbf{0} = [0]_{i=1}^n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

A.10 Function: $f : X \rightarrow Y$

Let X, Y be sets, and $f \subset X \times Y$. This f is called a function if and only if for all $x \in X$, if there exists $y, z \in Y$ such that $(x, y) \in f$ and $(x, z) \in f$, then $y = z$. In other words, a function associates either no element from Y to an element of X , or exactly one, but never two or more.

Notation: $f : X \rightarrow Y$ — f is a function that maps some or all elements of X to Y .

Note: a function which maps elements to natural numbers is called a sequence, e.g. $f : \mathbb{N}^+ \rightarrow \mathbb{R}$.

A.10.1 Value of a function at a point: $f(x)$

If for a given $x \in X$, a $y \in Y$ exists such that $(x, y) \in f$, then y is called the value of f at x .

Notation: $f(x) = y$.

A.10.2 Domain of a function: $\mathcal{D}_f$

$\mathcal{D}_f$ denotes the domain of the f function, ie. the set which contains all elements for which f is defined and nothing else:

$$\mathcal{D}_f = \{x : x \in X \text{ and there exists } y \in Y \text{ so that } (x, y) \in f\}$$

A.10.3 Image of a function: $\mathcal{R}_f$

$\mathcal{R}_f$ denotes the image of the function, ie. the set which contains all elements that are mapped to some $x \in \mathcal{D}_f$ and nothing else. Formally:

$$\mathcal{R}_f = \{f(x) : x \in \mathcal{D}_f\}$$

A.10.4 Multivariable, real-valued function: $f : \mathbb{R}^n \rightarrow \mathbb{R}$

For $n \in \mathbb{N}^+$, let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function. Such a function maps n -tuples of real numbers to a single real number.

A.10.5 Value of a multivariable, real-valued function at a point: $f(\mathbf{x}), f(x_1, x_2, \dots, x_n)$

If for a given $\mathbf{x} \in \mathbb{R}^n$, a $y \in \mathbb{R}$ exists such that $(x_1, x_2, \dots, x_n, y) \in f$, then y is called the value of f at $\mathbf{x}$.

Notation: $f(\mathbf{x}) = f(x_1, x_2, \dots, x_n) = y$.

A.11 Identity function: id

$\text{id} : \mathbb{R} \rightarrow \mathbb{R}$ — id is a function that maps real numbers to themselves:

$$\text{id}(x) = x$$

A.12 Commutativity

For all $x, y \in \mathbb{R}$:

$$x + y = y + x$$

A.13 Summation: $\sum_{i=1}^n a_i$

For $n \in \mathbb{N}^+$ and $\mathbf{a} \in \mathbb{R}^n$:

$$\sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$$

By convention, if $n = 1$ then

$$\sum_{i=1}^1 a_i = a_1$$

A.14 Distributivity: $c \cdot \sum_{i=1}^n a_i$

For $n \in \mathbb{N}^+$ and $\mathbf{a} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

$$c \cdot \sum_{i=1}^n a_i = \sum_{i=1}^n c \cdot a_i$$

A.15 Product of many numbers: $\prod_{i=1}^n a_i$

For $n \in \mathbb{N}^+$ and $\mathbf{a} \in \mathbb{R}^n$:

$$\prod_{i=1}^n a_i = a_1 \cdot a_2 \cdot \dots \cdot a_n$$

By convention, if $n = 1$ then

$$\prod_{i=1}^1 a_i = a_1$$

A.16 Sum and scaling of vectors: $\alpha \cdot \mathbf{a} + \beta \cdot \mathbf{b}$

For $n \in \mathbb{N}^+$ and $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, and $\alpha, \beta \in \mathbb{R}$:

$$\alpha \cdot \mathbf{a} + \beta \cdot \mathbf{b} = [\alpha \cdot a_i + \beta \cdot b_i]_{i=1}^n = \begin{bmatrix} \alpha \cdot a_1 & + & \beta \cdot b_1 \\ \alpha \cdot a_2 & + & \beta \cdot b_2 \\ & & \vdots \\ \alpha \cdot a_n & + & \beta \cdot b_n \end{bmatrix}$$

A.17 Dot product of two vectors: $\mathbf{a} \cdot \mathbf{b}$

For $n \in \mathbb{N}^+$ and $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i \cdot b_i$$

A.18 Hadamard product of two vectors: $\mathbf{a} \odot \mathbf{b}$

For $n \in \mathbb{N}^+$ and $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$:

$$\mathbf{a} \odot \mathbf{b} = [a_i \cdot b_i]_{i=1}^n = \begin{bmatrix} a_1 & \cdot & b_1 \\ a_2 & \cdot & b_2 \\ \vdots & & \vdots \\ a_n & \cdot & b_n \end{bmatrix}$$

A.19 Matrix: $\mathbf{W} \in \mathbb{R}^{n \times m}$

For $n, m \in \mathbb{N}^+$ and $i, j \in \mathbb{N}^+, i \leq n, j \leq m$, $\mathbf{W} \in \mathbb{R}^{n \times m}$ is called a "matrix" which contains the numbers $w_{i,j} \in \mathbb{R}$. In other words, $\mathbf{W}$ is a list of lists of numbers, arranged in a table that has n rows and m columns:

$$\mathbf{W} = \begin{bmatrix} w_{1,1} & w_{1,2} & \dots & w_{1,m} \\ w_{2,1} & w_{2,2} & \dots & w_{2,m} \\ \vdots & \vdots & & \vdots \\ w_{n,1} & w_{n,2} & \dots & w_{n,m} \end{bmatrix}$$

Notes:

- Rows and columns may be indexed starting from 0 instead of 1 as well.
- Individual rows and columns of a matrix may be notated by replacing the varying index with an asterisk. $\mathbf{W}_{i,*}$ means the i -th row vector of the $\mathbf{W}$ matrix, and $\mathbf{W}_{*,j}$ means the j -th column vector.
- When $n = 1$, then $\mathbf{W}$ may be called a "row vector" (since the matrix has only one row), and when $m = 1$, then $\mathbf{W}$ may be called a "column vector" (since the matrix has only one column).
- An $\mathbf{x} \in \mathbb{R}^n$ vector can be thought of as a matrix from $\mathbb{R}^{n \times 1}$.
- $\mathbf{0} \in \mathbb{R}^{n \times m}$ denotes the matrix in which all components are zero, and is called a "null matrix" or a "zero matrix". (The difference between a null vector and a null matrix is usually clear from the context.)

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

A.20 Transpose of a matrix: $\mathbf{W}^T$

Let $\mathbf{W} \in \mathbb{R}^{n \times m}$ be a matrix for $n, m \in \mathbb{N}^+$. The transpose of $\mathbf{W}$ is the $\mathbf{W}^T \in \mathbb{R}^{m \times n}$ matrix, where rows and columns switch roles like this:

$$\mathbf{W}^T = \begin{bmatrix} w_{1,1} & w_{2,1} & \dots & w_{n,1} \\ w_{1,2} & w_{2,2} & \dots & w_{n,2} \\ \vdots & \vdots & & \vdots \\ w_{1,n} & w_{2,n} & \dots & w_{n,n} \end{bmatrix}$$

A.21 Product of a matrix and a vector: $\mathbf{W} \cdot \mathbf{a}$

For $n, m \in \mathbb{N}^+$ and $\mathbf{W} \in \mathbb{R}^{n \times m}$ and $\mathbf{a} \in \mathbb{R}^m$, the product of the $\mathbf{W}$ matrix and the $\mathbf{a}$ vector is the following vector ($\mathbf{W} \cdot \mathbf{a} \in \mathbb{R}^n$):

$$\mathbf{W} \cdot \mathbf{a} = \left[\sum_{j=1}^m w_{i,j} \cdot a_j \right]_{i=1}^n = \begin{bmatrix} w_{1,1} \cdot a_1 + w_{1,2} \cdot a_2 + \dots + w_{1,m} \cdot a_m \\ w_{2,1} \cdot a_1 + w_{2,2} \cdot a_2 + \dots + w_{2,m} \cdot a_m \\ \vdots \\ w_{n,1} \cdot a_1 + w_{n,2} \cdot a_2 + \dots + w_{n,m} \cdot a_m \end{bmatrix}$$

Note: multiplying a matrix and a vector can have geometrical interpretations, e.g. rotations, reflections, and other linear transformations of a point in a space are often represented as a matrix-vector multiplication.

A.22 Sum and scaling of matrices: $\alpha \cdot \mathbf{A} + \beta \cdot \mathbf{B}$

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times m}$ be two matrices for $m, n \in \mathbb{N}^+$, and let $\alpha, \beta \in \mathbb{R}$ be two real numbers.

$$\alpha \cdot \mathbf{A} + \beta \cdot \mathbf{B} = \begin{bmatrix} \alpha \cdot a_{1,1} + \beta \cdot b_{1,1}, & \alpha \cdot a_{1,2} + \beta \cdot b_{1,2}, & \dots, & \alpha \cdot a_{1,m} + \beta \cdot b_{1,m} \\ \alpha \cdot a_{2,1} + \beta \cdot b_{2,1}, & \alpha \cdot a_{2,2} + \beta \cdot b_{2,2}, & \dots, & \alpha \cdot a_{2,m} + \beta \cdot b_{2,m} \\ \vdots & \vdots & & \vdots \\ \alpha \cdot a_{n,1} + \beta \cdot b_{n,1}, & \alpha \cdot a_{n,2} + \beta \cdot b_{n,2}, & \dots, & \alpha \cdot a_{n,m} + \beta \cdot b_{n,m} \end{bmatrix}$$

A.23 Matrix multiplication: $\mathbf{AB}$

For $n, m, p \in \mathbb{N}^+$, let $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times p}$ be two matrices. The product of $\mathbf{A}$ and $\mathbf{B}$ is the $\mathbf{AB} \in \mathbb{R}^{n \times p}$ matrix where the i -th element in the j -th column (for $n \geq i \in \mathbb{N}^+$ and $p \geq j \in \mathbb{N}^+$) is the dot product of the i -th row of $\mathbf{A}$ and the j -th column of $\mathbf{B}$:

$$\begin{aligned} \mathbf{AB} &= \begin{bmatrix} \mathbf{A}_{1,*} \cdot \mathbf{B}_{*,1}, & \mathbf{A}_{1,*} \cdot \mathbf{B}_{*,2}, & \dots, & \mathbf{A}_{1,*} \cdot \mathbf{B}_{*,p}, \\ \mathbf{A}_{2,*} \cdot \mathbf{B}_{*,1}, & \mathbf{A}_{2,*} \cdot \mathbf{B}_{*,2}, & \dots, & \mathbf{A}_{2,*} \cdot \mathbf{B}_{*,p}, \\ \vdots & \vdots & & \vdots \\ \mathbf{A}_{n,*} \cdot \mathbf{B}_{*,1}, & \mathbf{A}_{n,*} \cdot \mathbf{B}_{*,2}, & \dots, & \mathbf{A}_{n,*} \cdot \mathbf{B}_{*,p} \end{bmatrix} \\ &= \begin{bmatrix} \sum_{k=1}^m a_{1,k} \cdot b_{k,1}, & \sum_{k=1}^m a_{1,k} \cdot b_{k,2}, & \dots, & \sum_{k=1}^m a_{1,k} \cdot b_{k,p} \\ \sum_{k=1}^m a_{2,k} \cdot b_{k,1}, & \sum_{k=1}^m a_{2,k} \cdot b_{k,2}, & \dots, & \sum_{k=1}^m a_{2,k} \cdot b_{k,p} \\ \vdots & \vdots & & \vdots \\ \sum_{k=1}^m a_{n,k} \cdot b_{k,1}, & \sum_{k=1}^m a_{n,k} \cdot b_{k,2}, & \dots, & \sum_{k=1}^m a_{n,k} \cdot b_{k,p} \end{bmatrix} \end{aligned}$$

A.24 Hadamard product of two matrices: $\mathbf{A} \odot \mathbf{B}$

For $n, m \in \mathbb{N}^+$, let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times m}$ be two matrices of the same dimensions. The Hadamard product of $\mathbf{A}$ and $\mathbf{B}$ is the $\mathbf{A} \odot \mathbf{B} \in \mathbb{R}^{n \times m}$ matrix which contains the element-wise product of the elements of the two matrices:

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} a_{1,1} \cdot b_{1,1}, & a_{1,2} \cdot b_{1,2}, & \dots, & a_{1,m} \cdot b_{1,m} \\ a_{2,1} \cdot b_{2,1}, & a_{2,2} \cdot b_{2,2}, & \dots, & a_{2,m} \cdot b_{2,m} \\ \vdots & \vdots & & \vdots \\ a_{n,1} \cdot b_{n,1}, & a_{n,2} \cdot b_{n,2}, & \dots, & a_{n,m} \cdot b_{n,m} \end{bmatrix}$$

A.25 Square of a number: x^2

For $x \in \mathbb{R}$, $x^2 = x \cdot x$ is called "x squared", ie. the area of a square that has a side length of x .

A.26 Square of a function: $f^2(x)$

Let $f \in \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then $f^2 \in \mathbb{R} \rightarrow \mathbb{R}$ is also function, and for all $x \in \mathcal{D}_f$:

$$f^2(x) = (f(x))^2$$

A.27 Square root of a number: $\sqrt{x}$, $x^{\frac{1}{2}}$

For $x \in \mathbb{R}$, $\sqrt{x}$ is called the "square root of x ", ie. the numbers which give x when they are multiplied by themselves. In this document, we only care about the cases where $x \geq 0$, and we only consider the non-negative square root.

Note: $\sqrt{x}$ can also be written as $x^{\frac{1}{2}}$.

A.28 Absolute value: $|x|$

For $x \in \mathbb{R}$:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

A.29 Norm of a vector: $\|\mathbf{x}\|$

For $n \in \mathbb{N}^+$, if $\delta \in \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued function, then δ is a norm if it has all of the following properties for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- $\delta(\mathbf{x}) \geq 0$.
- $\delta(\mathbf{x}) = 0$ if and only if $x_j = 0$ for all $n \geq j \in \mathbb{N}^+$.
- $\delta(\alpha \cdot \mathbf{x}) = |\alpha| \cdot \delta(\mathbf{x})$.
- Triangle inequality: $\delta(\mathbf{x} + \mathbf{y}) \leq \delta(\mathbf{x}) + \delta(\mathbf{y})$.

Note: a norm can also be called the "length" of a vector, and for a given δ norm, $\delta(\mathbf{x} - \mathbf{y})$ can be called the "distance" of $\mathbf{x}$ and $\mathbf{y}$.

A.29.1 L_1 norm: $\|\mathbf{x}\|_1$

For $n \in \mathbb{N}^+$ and $\mathbf{x} \in \mathbb{R}^n$:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$$

A.29.2 L_2 norm, Euclidean norm: $\|\mathbf{x}\|_2$

For $n \in \mathbb{N}^+$ and $\mathbf{x} \in \mathbb{R}^n$:

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

Note: without context, the "length" of a vector usually means its Euclidean norm.

A.30 Minimum and maximum of a function

If $n \in \mathbb{N}^+$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function, and $\mathbf{a} \in \mathcal{D}_f$, then:

A.30.1 Global minimum

If for all $\mathbf{x} \in \mathcal{D}_f$, the $f(\mathbf{a}) \leq f(\mathbf{x})$ inequality holds, then f has a global minimum at $\mathbf{a}$.

A.30.2 Global maximum

If for all $\mathbf{x} \in \mathcal{D}_f$, the $f(\mathbf{a}) \geq f(\mathbf{x})$ inequality holds, then f has a global maximum at $\mathbf{a}$.

A.30.3 Local minimum

If there exists a $0 < \varepsilon \in \mathbb{R}$ such that for all $\mathbf{x} \in \mathcal{D}_f$ where the $\|\mathbf{a} - \mathbf{x}\| < \varepsilon$ inequality holds, $f(\mathbf{a}) \leq f(\mathbf{x})$, then f has a local minimum at $\mathbf{a}$.

In other words: if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{D}_f$ that is "close to" (but not equal to) $\mathbf{a}$, then f has a local minimum at $\mathbf{a}$.

Note: a global minimum is also a local minimum.

A.30.4 Local maximum

If there exists a $0 < \varepsilon \in \mathbb{R}$ such that for all $\mathbf{x} \in \mathcal{D}_f$ where the $\|\mathbf{a} - \mathbf{x}\| < \varepsilon$ inequality holds, $f(\mathbf{a}) \geq f(\mathbf{x})$, then f has a local maximum at $\mathbf{a}$.

In other words: if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{D}_f$ that is "close to" (but not equal to) $\mathbf{a}$, then f has a local maximum at $\mathbf{a}$.

Note: a global maximum is also a local maximum.

A.31 Integer power of a number: x^k

For $x \in \mathbb{R}$ and $k \in \mathbb{N}^+$ the k -th power of x :

$$x^k = \prod_{i=1}^k x$$

And the $-k$ -th power of x if $x \neq 0$:

$$x^{-k} = \frac{1}{x^k}$$

By convention, $x^0 = 1$.

A.32 Factorial: $n!$

For $n \in \mathbb{N}^+$, the factorial of n is the product of the natural numbers between 1 and n :

$$n! = \prod_{i=1}^n i = 1 \cdot 2 \cdot \dots \cdot n$$

By convention, $0! = 1$.

A.33 Limit of a single variable real-valued function: $\lim_{x \rightarrow c} f(x)$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function, and $c, L \in \mathbb{R}$ be real numbers. The limit of f at the point c is L if and only if for all $0 < \varepsilon \in \mathbb{R}$, there exists $0 < \delta \in \mathbb{R}$ such that for any $x \in \mathbb{R}$ which satisfies the $0 < |x - c| < \delta$ inequality, it holds that $|f(x) - L| < \varepsilon$.

(Less precisely: the limit of f at c is L if and only if f maps all numbers that are close to c to a number which is close to L .)

Notation:

$$\lim_{x \rightarrow c} f(x) = L$$

A.34 Limit of a real number sequence: $\lim_{n \rightarrow \infty} a_i$

The $a : \mathbb{N}^+ \rightarrow \mathbb{R}$ function defines a sequence of real numbers. Instead of $a(1), a(2), a(3), \dots$, a sequence is usually written as $(a_1, a_2, a_3, \dots)$.

If $\mathcal{D}_a = \mathbb{N}^+$, and an $L \in \mathbb{R}$ exists such that for any $0 < \varepsilon \in \mathbb{R}$, there exists $N \in \mathbb{N}^+$ so that for all $N < n \in \mathbb{N}^+$, the $|a_n - L| < \varepsilon$ inequality holds, then the sequence is called "convergent", and L is said to be its limit. Otherwise the sequence is called "divergent".

Notation:

$$\lim_{n \rightarrow \infty} a_i = L$$

A.35 Sum of infinite series: $\sum_{i=1}^{\infty} a_i$

Let $a : \mathbb{N}^+ \rightarrow \mathbb{R}$ be a sequence for which $\mathcal{D}_a = \mathbb{N}^+$. The following summation is called an "infinite series":

$$\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + \dots$$

For a given $n \in \mathbb{N}^+$, the n -th partial sum of the series is the sum of the first n elements of the sequence:

$$\sum_{i=1}^n a_i$$

If the limit of the n -th partial sums exists as n tends to infinity, then this limit is said to be the sum of the series, and the series is called "convergent":

$$\sum_{i=1}^{\infty} a_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i$$

Otherwise the series is called "divergent".

Note: the definition of convergence means that above a certain index, all elements in the series are closer to the limit than a certain error threshold $0 < \varepsilon \in \mathbb{R}$. Since practical computational problems rarely require precision above a certain error threshold (and computers with their finite memories are unable to represent arbitrary real numbers without some non-zero error anyway), when the value of a constant or a function is needed that is specified in terms of an infinite series, then it's usually enough to calculate a partial sum for a sufficiently large $n \in \mathbb{N}^+$ index which brings the error below the desired threshold.

A.36 Euler's constant: e

$$e = \sum_{i=0}^{\infty} \frac{1}{n!} = \frac{1}{1} + \frac{1}{1} + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \dots \approx 2.71828$$

A.37 Exponential function: e^x

For $x \in \mathbb{R}$:

$$\exp(x) = e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Note: for all $x, y \in \mathbb{R}$:

$$\begin{aligned} e^x \cdot e^y &= e^{x+y} \\ (e^x)^y &= e^{x \cdot y} \end{aligned}$$

A.38 Natural logarithm: $\ln(x)$

For $0 < x \in \mathbb{R}$, $\ln(x)$ is the natural logarithm of x , which is the inverse of the exponential function:

$$e^{\ln(x)} = x$$

A.39 Real power of a number: α^x

For $x \in \mathbb{R}$ and $0 < \alpha \in \mathbb{R}$:

$$\alpha^x = \left(e^{\ln(\alpha)} \right)^x = e^{\ln(\alpha) \cdot x}$$

Note: for all $x, y \in \mathbb{R}$:

$$\begin{aligned} \alpha^x \cdot \alpha^y &= \alpha^{x+y} \\ (\alpha^x)^y &= \alpha^{x \cdot y} \\ \alpha^{\frac{1}{2}} &= \sqrt{\alpha} \end{aligned}$$

A.40 Differentiation**A.40.1 Differentiation of single variable, real-valued functions: $f'(a)$, $\frac{d}{dx}f(a)$, $\frac{df}{dx}(a)$, f' , $\frac{d}{dx}f$, $\frac{df}{dx}$**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function and $a \in \mathcal{D}_f$. The f function is differentiable at the point a if an $I \subset \mathcal{D}_f$ non-empty open interval and an $L \in \mathbb{R}$ number exist such that $a \in I$ and

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = L$$

In other words, f is differentiable if for all $0 < \varepsilon \in \mathbb{R}$, there exists $0 < \delta \in \mathbb{R}$ such that for every $0 \neq h \in \mathbb{R}$ which satisfies the $|h| < \delta$ inequality, $a+h \in \mathcal{D}_f$, and

$$\left| L - \frac{f(a+h) - f(a)}{h} \right| < \varepsilon$$

If f is differentiable at a , ie. the limit L exists, then L is called the "derivative of f at a ", and it is denoted as $f'(a) = L$ (read: "f prime of a").

Alternative notations (Leibniz):

$$f'(a) = \frac{df}{dx}(a) = \frac{d}{dx}f(a)$$

If f is differentiable for all $a \in \mathcal{D}_f$, then f is differentiable, and its derivative is f' (or $\frac{d}{dx}f$).

An example of a function which is not differentiable on the entirety of its domain is the $\text{abs} : \mathbb{R} \rightarrow \mathbb{R}$, $\text{abs}(x) = |x|$ function, because it is not differentiable at $a = 0$: the $\frac{\text{abs}(0+h) - \text{abs}(0)}{h}$ expression yields -1 for all $0 > h \in \mathbb{R}$, but it yields $+1$ for all $0 < h \in \mathbb{R}$, therefore the single limit that the definition requires does not exist at 0. However, $\text{abs}(x)$ is differentiable everywhere on both the $(-\infty, 0)$ and the $(0, +\infty)$ intervals.

A.40.1.1 Practical applications There are several practical applications of the derivative. Examples include:

- **Momentary speed:** in physics, if the distance that is covered by some object from a starting point during $t \in \mathbb{R}$ seconds is given by the $s : \mathbb{R} \rightarrow \mathbb{R}$ function, then the average velocity for this journey can be calculated as $v = \frac{s(t)}{t}$, but the momentary velocity (that is shown on the object's speed meter at a given moment) is given by $s'(t)$.
- **Optimization:** $f'(a)$ tells the slope of the tangent line to the graph of f at the point a , as shown on figure 13 in A.40.1.3. In other words, $f'(a)$ tells how fast the function increases or decreases near the point a . This is useful for finding local minima and maxima of a function (in other words: to optimize the thing that is modeled by the function), because at such points, the derivative of the function is 0. However, generally it doesn't necessarily mean that whenever the derivative is 0, then there's a local minimum or maximum — to find out if a point is a local extremum, further tests need to be performed, e.g. by checking the second derivative (which is the derivative of the derivative if that exists). Finding the global minima and maxima can often be done by finding all the local minima and maxima, and then selecting the global extrema among those.

A.40.1.2 Properties Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be two differentiable functions and let $\alpha, \beta \in \mathbb{R}$ be two real numbers, and let $z \in \mathbb{Z}$ be an integer.

A.40.1.2.1 Derivative of constant If for all $x \in \mathcal{D}_f$ it holds that $f(x) = \alpha$, then $f'(x) = 0$.

A.40.1.2.2 Derivatives of powers: $(x^z)'$ For all $x \in \mathbb{R}$:

$$(x^z)' = z \cdot x^{z-1}$$

A.40.1.2.3 Derivative of e^x For all $x \in \mathbb{R}$:

$$(e^x)' = e^x$$

A.40.1.2.4 Derivative of α^x If $\alpha > 0$, then for all $x \in \mathbb{R}$:

$$(\alpha^x)' = \alpha^x \cdot \ln(\alpha)$$

A.40.1.2.5 Derivative of $\ln(x)$ For all $0 < x \in \mathbb{R}$:

$$\ln(x)' = \frac{1}{x}$$

A.40.1.2.6 Linearity: $(\alpha \cdot f(x) \pm \beta \cdot g(x))'$ For all $x \in \mathcal{D}_f \cap \mathcal{D}_g$:

$$(\alpha \cdot f(x) \pm \beta \cdot g(x))' = \alpha \cdot f'(x) \pm \beta \cdot g'(x)$$

A.40.1.2.7 Product rule: $(f(x) \cdot g(x))'$ For all $x \in \mathcal{D}_f \cap \mathcal{D}_g$:

$$(f(x) \cdot g(x))' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

A.40.1.2.8 Quotient rule: $\left(\frac{f(x)}{g(x)}\right)'$ For all $x \in \mathcal{D}_f \cap \mathcal{D}_g$ where $g(x) \neq 0$:

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}$$

A.40.1.2.9 Chain rule: $(f(g(x)))'$ For all $x \in \mathcal{D}_g$ where $g(x) \in \mathcal{D}_f$:

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

A.40.1.3 Example Consider the following function:

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = (x+1)^3 - (x-1)^2 - 12 \cdot x$$

Its graph is shown on figure 13.

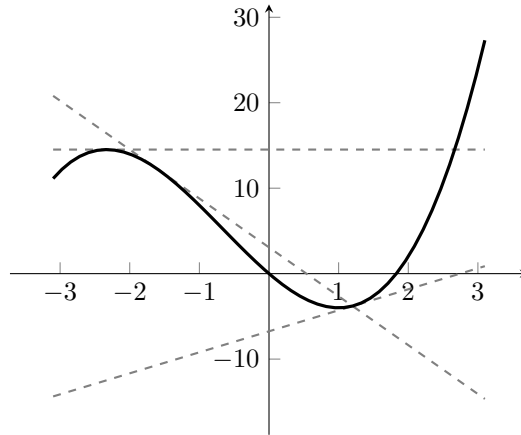


Figure 13: Graph of the $f(x) = (x+1)^3 - (x-1)^2 - 12 \cdot x$ function from example A.40.1.3, with its tangent lines at $x = -1.6$, at $x = -\frac{7}{3}$, and at $x = 1.23$. The slopes of the tangent lines are given by $f'(-1.6)$, $f'(-\frac{7}{3})$, and $f'(1.23)$ respectively. The function has a local maximum at $x = -\frac{7}{3}$, so the tangent line there is horizontal, ie. its slope is 0, ie. $f'(-\frac{7}{3}) = 0$.

This function is differentiable on the entire $\mathbb{R}$ (proof left for the reader). A step by step calculation of its derivative:

$$\begin{aligned}
f'(x) &= ((x+1)^3 - (x-1)^2 - 12 \cdot x)' \\
&= ((x+1)^3 - (x-1)^2)' - (12 \cdot x)' \\
&= ((x+1)^3)' - ((x-1)^2)' - (12 \cdot x)' \\
&= ((x+1)^3)' - ((x-1)^2)' - 12 \cdot (x)' \\
&= (3 \cdot (x+1)^2 \cdot (x+1)') - (2 \cdot (x-1) \cdot (x-1)') - 12 \cdot 1 \\
&= (3 \cdot (x+1)^2 \cdot ((x)' + (1)')) - (2 \cdot (x-1) \cdot ((x)' - (1)')) - 12 \\
&= (3 \cdot (x+1)^2 \cdot (1+0)) - (2 \cdot (x-1) \cdot (1-0)) - 12 \\
&= (3 \cdot (x+1)^2 \cdot 1) - (2 \cdot (x-1) \cdot 1) - 12 \\
&= 3 \cdot (x+1)^2 - 2 \cdot (x-1) - 12
\end{aligned}$$

A.40.2 Partial derivatives of multivariable, real-valued functions: $\frac{\partial}{\partial x_i} f(\mathbf{a})$, $\frac{\partial f}{\partial x_i}(\mathbf{a})$, $\frac{\partial}{\partial x_i} f$, $\frac{\partial f}{\partial x_i}$

For $n \in \mathbb{N}^+$, let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function which maps vectors from $\mathbb{R}^n$ to real numbers.

For $n \geq i \in \mathbb{N}^+$, the partial derivative of f in the direction x_i at $\mathbf{a}$ is the following limit:

$$\frac{\partial}{\partial x_i} f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(a_1, a_2, \dots, a_i + h, \dots, a_n) - f(a_1, a_2, \dots, a_i, \dots, a_n)}{h} \quad (30)$$

Note: $\frac{\partial}{\partial x_i} f(\mathbf{a})$ can also be written as $\frac{\partial f}{\partial x_i}(\mathbf{a})$, and $\frac{\partial f}{\partial x_i} : \mathbb{R}^n \rightarrow \mathbb{R}$ can be thought of as a function in its own right.

The practical use of partial derivatives is that $\frac{\partial}{\partial x_i} f(\mathbf{a})$ tells the slope of the tangent line to the graph of f at $\mathbf{a}$ along the axis that corresponds to the x_i variable, similarly to the single variable case. (Figure 14 in A.40.2.2 shows what this means for the $n = 2$ case.)

Since all the numbers in the expression in equation 30 are constants except for a_i , the expression can be treated just like a single variable function, and so the partial derivatives can be calculated the exact same way, with the exact same rules. In other words, if we define the $g : \mathbb{R} \rightarrow \mathbb{R}$ single variable function for a given $\mathbf{a} \in \mathbb{R}^n$ and $n \geq i \in \mathbb{N}^+$ as $g(x) = f(a_1, a_2, \dots, x, \dots, a_n)$ with x being the i -th parameter of f , then $\frac{\partial}{\partial x_i} f(\mathbf{a}) = g'(a_i)$.

A.40.2.1 Gradient vector: $\nabla f(\mathbf{a})$ For a given $\mathbf{a} \in \mathbb{R}^n$ vector, if all the partial derivatives $\frac{\partial}{\partial x_i}$ (for $n \geq i \in \mathbb{N}^+$) of f are defined at $\mathbf{a}$, then the gradient of f at $\mathbf{a}$ is the following vector ($\nabla f(\mathbf{a}) \in \mathbb{R}^n$):

$$\nabla f(\mathbf{a}) = \left[\frac{\partial}{\partial x_i} f(\mathbf{a}) \right]_{i=1}^n = \begin{bmatrix} \frac{\partial}{\partial x_1} f(\mathbf{a}) \\ \frac{\partial}{\partial x_2} f(\mathbf{a}) \\ \vdots \\ \frac{\partial}{\partial x_n} f(\mathbf{a}) \end{bmatrix}$$

A.40.2.2 Example Consider the following function:

$$\begin{aligned}
f : \mathbb{R}^2 &\rightarrow \mathbb{R} \\
f(x_1, x_2) &= \frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot x_2^2) + 3
\end{aligned}$$

Its graph is a 3-dimensional surface, as shown on figure 14, because this function can be thought of as if it assigned a height value for each point of a 2-dimensional plane.

For example, the height of the surface above the $\mathbf{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \in \mathbb{R}^2$ point on the plane is $f(3, 1) = \frac{1}{50} \cdot ((3 - 1)^3 - 5 \cdot 1^2) + 3 = 3.06$.

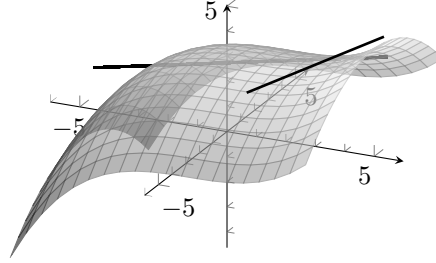


Figure 14: Graph of the $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = \frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot x_2^2) + 3$ function, and its tangent lines at $\mathbf{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, parallel with the x_1 and x_2 axis, as given by $\nabla f(\mathbf{a})$.

The partial derivatives exist for both of its variables. (Proof left for the reader.)

If we slice this surface along the $x_2 = 1$ line which lies on the plane (this line is parallel to the x_1 axis), then we get the first graph that is shown on figure 15, and $\frac{\partial}{\partial x_1} f(3, 1)$ will tell the slope of the tangent line to this graph at the $x_1 = 3$ point. In other words, if we define the $g : \mathbb{R} \rightarrow \mathbb{R}$ function as $g(x_1) = f(x_1, a_2) = \frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot 1^2) + 3$, then g can be differentiated as a single variable function, and $\frac{\partial}{\partial x_1} f(\mathbf{a}) = g'(a_1)$.

Similarly, slicing this function along the $x_1 = 3$ line (which is parallel to the x_2 axis) gives the second graph that is shown on figure 15, and $\frac{\partial}{\partial x_2} f(3, 1)$ will tell the slope of the tangent line to this graph at the $x_2 = 1$ point. In other words, if we define the $h : \mathbb{R} \rightarrow \mathbb{R}$ function as $h(x_2) = f(a_1, x_2) = \frac{1}{50} \cdot ((3 - 1)^3 - 5 \cdot x_2^2) + 3$, then h can be differentiated as a single variable function as well, and $\frac{\partial}{\partial x_2} f(\mathbf{a}) = h'(a_2)$.

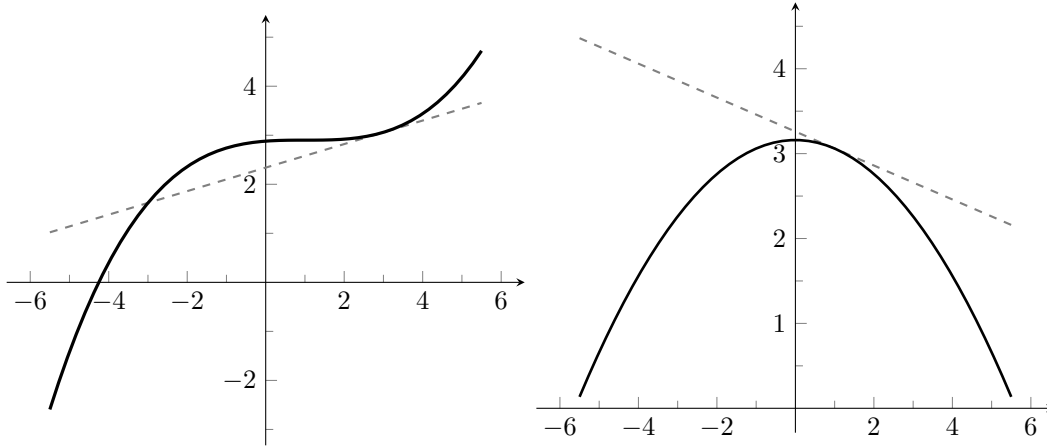


Figure 15: Graphs of the $g(x_1) = f(x_1, a_2) = \frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot 1^2) + 3$ and the $h(x_2) = f(a_1, x_2) = \frac{1}{50} \cdot ((3 - 1)^3 - 5 \cdot x_2^2) + 3$ slices of the f function, and the tangent lines of these slices at $x_1 = 3$ and at $x_2 = 1$, as given by $\frac{\partial}{\partial x_1} f(3, 1)$ and $\frac{\partial}{\partial x_2} f(3, 1)$ respectively.

Below is the calculation of the partial derivatives for the $\mathbf{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ point, step by step. We are going to use the single value differentiation rules, and the fact that if $\alpha \in \mathbb{R}$ is a constant, then $(\alpha - 1)^3$ and $5 \cdot \alpha^2$ are also constants, therefore their derivatives are 0.

$$\begin{aligned}
\frac{\partial}{\partial x_1} f(x_1, a_2) &= \left(\frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot a_2^2) + 3 \right)' \\
&= \left(\frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot a_2^2) \right)' + (3)' \\
&= \left(\frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot a_2^2) \right)' + 0 \\
&= \left(\frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot a_2^2) \right)' \\
&= \frac{1}{50} \cdot ((x_1 - 1)^3 - 5 \cdot a_2^2)' \\
&= \frac{1}{50} \cdot (((x_1 - 1)^3)' - (5 \cdot a_2^2)') \\
&= \frac{1}{50} \cdot (((x_1 - 1)^3)' - 0) \\
&= \frac{1}{50} \cdot ((x_1 - 1)^3)' \\
&= \frac{1}{50} \cdot (3 \cdot (x_1 - 1)^2 \cdot (x_1 - 1)') \\
&= \frac{1}{50} \cdot (3 \cdot (x_1 - 1)^2 \cdot ((x_1)' - (1)')) \\
&= \frac{1}{50} \cdot (3 \cdot (x_1 - 1)^2 \cdot (1 - 0)) \\
&= \frac{1}{50} \cdot (3 \cdot (x_1 - 1)^2 \cdot 1) \\
&= \frac{3}{50} \cdot (x_1 - 1)^2
\end{aligned}$$

Now $\frac{\partial}{\partial x_2} f(a_1, x_2)$ with somewhat bigger steps:

$$\begin{aligned}
\frac{\partial}{\partial x_2} f(a_1, x_2) &= \left(\frac{1}{50} \cdot ((a_1 - 1)^3 - 5 \cdot x_2^2) + 3 \right)' \\
&= \left(\frac{1}{50} \cdot ((a_1 - 1)^3 - 5 \cdot x_2^2) \right)' + (3)' \\
&= \frac{1}{50} \cdot ((a_1 - 1)^3 - 5 \cdot x_2^2)' + 0 \\
&= \frac{1}{50} \cdot (((a_1 - 1)^3)' - (5 \cdot x_2^2)') \\
&= \frac{1}{50} \cdot (0 - 5 \cdot (x_2^2)') \\
&= \frac{1}{50} \cdot (-5) \cdot (x_2^2)' \\
&= -\frac{1}{10} \cdot (2 \cdot x_2 \cdot (x_2)') \\
&= -\frac{1}{10} \cdot (2 \cdot x_2 \cdot 1) \\
&= -\frac{1}{5} \cdot x_2
\end{aligned}$$

Therefore the gradient vector at the $\mathbf{a} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \in \mathbb{R}^2$ point is:

$$\nabla f(\mathbf{a}) = \begin{bmatrix} \frac{\partial}{\partial x_1} f(3, 1) \\ \frac{\partial}{\partial x_2} f(3, 1) \end{bmatrix} = \begin{bmatrix} \frac{3}{50} \cdot (3 - 1)^2 \\ -\frac{1}{5} \cdot 1 \end{bmatrix} = \begin{bmatrix} \frac{6}{25} \\ -\frac{1}{5} \end{bmatrix}$$

Note: the math works the same way for more than 2 dimensions ($n > 2$), it's just harder to visualize.